Projective cohomology and smooth affine models

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Course sources: https://github.com/KokunoYumeto/open-math-courses/tree/main/docs/courses/AG-RG.

Contents

Projective cohomology and smooth affine models **1**

1. Earlier programme proofs used in this lesson 1
2. Formal functions on a projective scheme 4
3. Smoothness and line bundles at a finite stage 6
4. Geometrically connected smooth affine models 10
5. A universal finite complex for a line bundle 18
6. The smooth curve regular immersion calculation 19
7. Applying the proof route in the group lessons 28

Freely readable sources and history 28

History 28

Projective cohomology and smooth affine models

This supporting lesson proves the projective formal functions and smooth model statements needed before the reductive group lessons. It also records the precise earlier programme proofs used for completion, projective finiteness and curve duality. It follows the supporting lesson Algebra and sheaf cohomology before reductive groups, whose Sections 1–9 supply the affine, homological, projective-space and Serre calculations. Both supporting lessons precede AG-RG-01. All mathematical sources listed here are freely readable. Source citations identify the material used to write the proofs; they do not replace any proof below.

The formal functions exposition follows the freely readable Stacks Project proofs. The smooth affine component argument below specializes the Stacks model route and supplies its intermediate arguments. The text of this lesson is dedicated to the public domain under CC0 1.0; the Stacks Project itself is distributed under the GNU FDL 1.2 or later.

1. Earlier programme proofs used in this lesson

The following references are to proofs already written in programme lessons. Their numbered statements delimit exactly what is used.

- **AG-CA-03, Noetherian and Artinian rings:** Proposition 1.2 (finite modules over a Noetherian ring are Noetherian), Theorem 2.1 (Hilbert basis), Theorem 5.1 Artin–Rees, and Theorem 6.1 (Krull intersection). In particular Artin–Rees is proved by finite generation of the graded submodule of the Rees module; it is not taken from a book.
- **AG-CA-19, Completion:** Theorem 1.2 (exact inverse limits with a Mittag–Leffler kernel), Lemma 2.1 (completion preserves surjections with the induced kernel filtration), Proposition 2.2 (finite ideal powers in a completion), Theorems 3.1–3.3 (finite-module exactness, completion tensor comparison, flatness, faithful flatness and Noetherianity). These statements have full proofs there. Thus a finite module over an already complete Noetherian ring is complete, and local Noetherian completion is faithfully flat.
- **AG-CA-17, Formally smooth, unramified and étale ring maps:** Theorem 3.1 (formal smoothness is the split conormal criterion), Theorems 4.1 and 5.1 (standard smooth presentations and local standard form), and Lemma 6.2 (unique lifting of idempotents across a square-zero ideal). These are written lifting and matrix proofs. Lemma 1.1 extends idempotent lifting to a nilpotent ideal by successive square-zero steps.
- **AG-CA-18, Smooth algebras over a field and the Jacobian criterion:** Theorem 2.1 proves regularity of smooth field-algebra local rings. Its proof uses standard smooth equations and the regular parameter quotient theorem. Base change of smoothness, proved in AG-CA-17, consequently gives geometric regularity.
- **AG-CA-14, Regular local rings:** Theorem 1.1 and Proposition 1.4 prove regular parameter quotients and generation of a regular quotient's ideal by part of a regular system of parameters.

The graded and regular-sequence assertions used there are the full proof of Theorem 6.1 in **Regular sequences, depth and Cohen–Macaulay modules**. The Koszul resolution and its Hom calculation are the corresponding earlier depth lesson proofs.

- **AG-CA-07, Tor and flat modules:** Theorems 5.3 and 6.1 contain the finite-projective and flat-kernel calculations corresponding to Section 5. That section writes its universal-tail flatness and global splitting arguments directly, without taking the Tor criterion as an additional prerequisite.
- **AG-CA-05, Integral extensions:** Theorems 3.2–3.4 prove lying over and incomparability. Section 6 below contains the complete DVR argument in Lemma 6.C and the separable trace-lattice, finite normalization, function-field rational-map spreading and dense equalizer arguments in Lemma 6.4. The homogeneous-coordinate extension over every DVR is written there as well.
- **AG-MO-03, Limits of schemes and Noetherian approximation:** Theorem 1.1, Lemma 2.1 and Theorems 4.1–4.2 prove affine limits, retention of finite open covers, finite-presentation descent, eventual equality of maps and descent of closed immersions. Only the proved affine approximation Theorem 5.1 is needed here. The external-proof-only absolute approximation Theorem 5.2 is not used.
- **AG-MO-06, Quasi-finite morphisms and Chevalley's theorem:** Theorem 3.1 proves the Noetherian constructibility criterion and Theorem 4.2 proves Chevalley's theorem. The argument in Section 4 below uses these particular proved statements.
- **AG-FSE, Flat morphisms:** Lemma 5.1 and Corollary 5.2 prove generic freeness over a domain, including for a finite module over a finite type algebra. No openness-of-flatness theorem is needed in Section 4.
- **AG-CA-18, Smooth algebras over a field and the Jacobian criterion:** Theorem 6.1 and Corollary 6.2 prove smooth and étale flatness over arbitrary bases. Together with AG-CA-17 Theorem 5.1 and Lemma 7.1 these give the separating generic coordinates used below.
- **AG-RG-S01, Sections 4–6:** the finite ordered Čech comparison, affine quasi-coherent vanishing and flat base change are proved before this lesson. A finite affine cover of a separated scheme therefore computes quasi-coherent cohomology by its finite section complex. The same written complex calculation treats finite and arbitrary direct sums and flat localization.
- **AG-RG-S01, Sections 7–8:** Theorems 7.2–7.3 compute every projective-space twist and the perfect monomial pairing. Proposition 8.2 and Theorem 8.4 prove finite quotients by twists, finite coherent cohomology and eventual vanishing, including the closed-immersion passage to projective schemes. Lemmas 8.1 and 8.3 give denominator extension and the ample-power immersion; Corollary 7.B makes it closed for a proper source. Theorem 8.5 treats every sufficiently large power, as applied in Section 7 below.
- **AG-QC-14, Ext sheaves and Serre duality on projective space:** Proposition 1.1, Proposition 1.2, Proposition 2.1 and Theorem 5.1 give the local Ext calculation, vector-bundle tensor-Hom comparison, local-to-global Ext spectral sequence and ambient projective Serre duality. The duality proof uses the monomial trace, a two-stage twist presentation and explicit dimension shifting. Its reference to "proper finiteness" in Theorem 4.1 is supplied here by the full projective finiteness proof in Algebra and sheaf cohomology before reductive groups, Section 8 (AG-QC-05 Theorem 2.1), without using the general proper-finiteness provider. Section 6 below also spells out the single-row local-to-global Ext comparison being consumed, using the earlier finite filtered-complex construction; no general proper duality theorem is needed here.

These references are a finite proof dependency list, not a declaration that unnamed results are ordinary foundations. In particular the further smoothness descent and connected model results are proved in Sections 3–4.

1.1. The consumed Artin–Rees and completion arguments

The following records the actual arguments of AG-CA-03 Theorem 5.1 and AG-CA-19 Theorems 3.1–3.3 before their uses. Let A be Noetherian and $I \subset A$ an ideal.

Lemma 1.1 (Artin–Rees). If $N \subset M$ are finite A-modules, there is c such that

$$N \cap I^n M = I^{n-c}(N \cap I^c M) \quad (n \geq c).$$

Proof. The Rees algebra $B = \bigoplus I^n t^n$ is generated by the finitely many generators of I in degree one, so is Noetherian by Hilbert basis. The module $\bigoplus I^n M t^n$ is finite over it, generated by any finite list of generators of M in degree zero. Its graded submodule $\bigoplus (N \cap I^n M) t^n$ is finite. Choose homogeneous generators of degrees at most c. For $n \geq c$, the degree n component is a sum of $I^{n-d}(N \cap I^d M)$, for $d \leq c$; each is contained in $I^{n-c}(N \cap I^c M)$. The reverse inclusion follows by multiplication and the submodule property. □

The consumed Krull intersection statement also follows from this proof. For a finite module over a Noetherian local ring, with I in its maximal ideal, the submodule $N = \bigcap_n I^n M$ is finite. Lemma 1.1 gives $N = IN$ by taking $n = c + 1$, since $N \cap I^n M = N$ at every index. Nakayama gives $N = 0$. This is the actual AG-CA-03 Theorem 6.1 argument used in the earlier regular-ring proof.

Lemma 1.2 (exact finite-module completion). Writing $\widehat{M} = \varprojlim M/I^n M$, completion is exact on finite modules and

$$\widehat{M} = M \otimes_A \widehat{A}.$$

Proof. For $0 \to N \to M \to L \to 0$, the quotients give

$$0 \to N/(N \cap I^n M) \to M/I^n M \to L/I^n L \to 0.$$

The kernel system has surjective transitions. Compatible lifts in the middle system can therefore be chosen successively: given a lift at stage n, take any lift of the desired next quotient element and adjust its image's difference at stage n by lifting that difference from the kernel system. This proves surjectivity on inverse limits. Injectivity and equality with the limit kernel hold coordinatewise. By Lemma 1.1, $I^n N \subset N \cap I^n M \subset I^{n-c} N$; hence the two filtrations of N are cofinal. Their limits identify explicitly by taking a sufficiently far coordinate to define a coordinate of the other system, independent of that choice by compatibility. Thus the displayed inverse-limit sequence is $0 \to \widehat{N} \to \widehat{M} \to \widehat{L} \to 0$.

A finite module has a finite presentation, since submodules of finite modules are finite over a Noetherian ring. Complete $A^r \to A^s \to M \to 0$, using the just-proved exactness. Tensor the same presentation with $\widehat{A}$, using tensor's right exactness. Finite direct sums complete coordinatewise, so both cokernels are the cokernel of the same matrix $\widehat{A}^r \to \widehat{A}^s$. This gives the canonical tensor comparison. In particular, when A is already complete, every finite module is complete. □

Lemma 1.3 (flat, faithfully flat and Noetherian completion). The ring $\widehat{A}$ is Noetherian, is complete for $I\widehat{A}$, and is A-flat. If I is contained in the Jacobson radical, the completion is faithfully flat.

Proof. Flatness follows directly from the preceding finite-module exactness. Every module is the filtered union of its finite generated submodules. A tensor element uses finitely many such generators, and an equality to zero holds in a sufficiently large finite submodule, because tensor commutes with filtered colimits by its generators and relations. For an injection $M' \to M$, put a proposed tensor kernel element in a finite submodule $N \subset M'$, and put its vanishing equality in some finite submodule of M containing the image of N. Exactness of completion for these finite modules proves injectivity there, hence kills the proposed kernel. Thus tensor with $\widehat{A}$ preserves every injection.

For faithfulness, if $M \neq 0$, choose $0 \neq x \in M$, with cyclic submodule $Ax = A/J$. Choose a maximal ideal $\mathfrak{m} \supset J$. Since $I \subset \mathfrak{m}$, the module $A/\mathfrak{m}$ is unchanged by completion. The tensor comparison gives $\widehat{A}/\mathfrak{m}\widehat{A} = A/\mathfrak{m} \neq 0$, a quotient of $(A/J) \otimes \widehat{A}$. Flatness injects the latter into $M \otimes \widehat{A}$, so this is nonzero. This proves faithful flatness and, by applying it to a tensor kernel, detection of exactness.

Apply Lemma 1.2 to $0 \to I^a \to A \to A/I^a \to 0$. The last module is unchanged by completion, since a power of I kills it. The image of $I^a \otimes \widehat{A}$ is exactly $I^a\widehat{A}$, whence

$$\widehat{A}/I^a\widehat{A} = A/I^a.$$

Writing $J = I\widehat{A}$, its powers are $I^a\widehat{A}$; these identities and the defining inverse limit prove J-adic completeness and separatedness. Their successive kernels identify $\operatorname{gr}_J \widehat{A}$ with $\operatorname{gr}_I A$. The latter is a quotient of a polynomial ring over A/I on the finite generators of I, and is Noetherian.

For any ideal $Q \subset \widehat{A}$, its initial classes form a homogeneous ideal of this Noetherian graded ring. Choose finite homogeneous generators and lift them to $q_i \in Q$, with orders d_i. For $q \in Q$, subtract at stage n a combination $\sum_i a_{i,n} q_i$ matching its residual initial class, with $a_{i,n} \in J^{n-d_i}$ and zero when $n < d_i$. The new residual is in J^{n+1}. Each coefficient series converges, since its orders tend to infinity. Its sums a_i satisfy $q = \sum_i a_i q_i$ by separatedness. Thus every ideal is finite and $\widehat{A}$ is Noetherian. No prior assertion that Q is closed was used. □

For a local ring with maximal ideal $\mathfrak{m}$, its completion is local: an element with nonzero residue is a unit in every quotient $A/\mathfrak{m}^n$, and the compatible inverses give its inverse in the limit. Projection onto the residue field is surjective by successive lifting. The preceding quotient identity makes its kernel $\mathfrak{m}\widehat{A}$. Thus Lemma 1.3 gives the exact local completion statement used by the group lessons.

Lemma 1.4 (idempotents across a nilpotent ideal). Every idempotent of R/J, for a nilpotent ideal J, has a unique idempotent lift to R.

Proof. First suppose $J^2 = 0$. For a lift e, put $\delta = e^2 - e \in J$. Then $e' = e - (2e-1)\delta$ is idempotent: expand its square and use $\delta^2 = 0$ and $(2e-1)^2 = 1 + 4\delta$. For two idempotent lifts the difference $u \in J$ satisfies $(2e-1)u = 0$; the element $2e-1$ is a unit since its square is one, so $u = 0$. Apply this square-zero argument successively to J^a/J^{a+1}, starting with $a = 1$; each is square-zero because $2a \geq a + 1$, and nilpotence makes the list finite. □

2. Formal functions on a projective scheme

Let A be Noetherian, $I \subset A$ an ideal, $X \to \operatorname{Spec} A$ projective and F coherent. Write

$$B = \bigoplus_{n \geq 0} I^n t^n \subset A[t], \qquad M_n^q = H^q(X, I^n F), \qquad M^q = M_0^q.$$

The transition maps $M_m^q \to M_n^q$, for $m \geq n$, come from inclusion of ideal powers. Multiplication $I^d \otimes M_n^q \to M_{n+d}^q$ is compatible with these transitions: both sheaf maps are multiplication followed by the same inclusion. Applying cohomology preserves their equality.

Lemma 2.1. The graded B-module $\bigoplus_{n \geq 0} M_n^q t^n$ is finite for every q.

Proof. Generators of I, placed in degree one, generate B as an A-algebra; Hilbert basis makes B Noetherian. Let $X_B = X \times_A \operatorname{Spec} B$, with affine projection π. On $U = \operatorname{Spec} R \subset X$, put $F|_U = \widetilde{N}$. The graded module $\bigoplus I^n N$ is finite over $R \otimes_A B$: finitely many R-generators of N, placed in degree zero, generate it. Here $R \otimes_A B \to \bigoplus I^n R$ is a surjection, and need not be an isomorphism. No flatness assumption is being made.

Localization gives the same modules on overlaps. They glue to a coherent sheaf G on X_B, with $\pi_* G = \bigoplus I^n F$. Coherence holds because $R \otimes_A B$ is Noetherian and the module is finite. Base

changing a projective closed immersion shows that X_B is projective over B; AG-QC-05 Theorem 2.1 and its closed-immersion passage give finite $H^q(X_B, G)$ over B.

Use the inverse images of a finite affine cover of X. Each intersection is affine on both schemes, and its sections of G are the sections of $\bigoplus I^n F$ on the original intersection. Thus their ordered section complexes are identical. Finite products in that complex commute with direct sums, and direct sums of modules are exact. Taking cohomology therefore identifies

$$H^q(X_B, G) = \bigoplus_{n \geq 0} H^q(X, I^n F),$$

with the graded B-action preserved. This proves finiteness. □

This writes the projective case of the free proofs Stacks 02O8 and 0897. General proper coherent pushforward and its dévissage are unnecessary: both spaces just used are projective.

Put

$$J_n^q = \operatorname{im}(M_n^q \to M^q), \qquad K_n^q = \ker(M_n^q \to M^q).$$

Lemma 2.2. There are constants $a_q, b_q \geq 0$ such that

$$I^n M^q \subset J_n^q \subset I^{n-a_q} M^q \quad (n \geq a_q),$$

and $K_m^q \to K_n^q$ is zero whenever $m \geq n + b_q$.

Proof. Multiplication by $u \in I^n$ on F factors through $I^n F$, proving the first inclusion. The image and kernel of the graded map from Lemma 2.1 to the corresponding degreewise copies of M^q are graded subquotients of a finite B-module. More explicitly, multiplication and transition compatibility show that $\bigoplus J_n^q t^n$ is the graded image module and $\bigoplus K_n^q t^n$ is a graded submodule of $\bigoplus M_n^q t^n$. Both are finite, because B is Noetherian.

Choose homogeneous generators of the image module of degrees $d \leq a_q$. Its degree n is a sum of terms $I^{n-d} J_d^q \subset I^{n-a_q} M^q$, proving the other inclusion. Choose homogeneous kernel generators $z_j \in K_{d_j}^q$, with $d_j \leq b_q$. An element in degree m is a sum of $u_j z_j$, where $u_j \in I^{m-d_j}$. If $m \geq n + b_q$, express u_j as a sum of products vw, $v \in I^n$, $w \in I^{m-d_j-n}$. Transition of vwz_j to M_n^q factors as transition of z_j to M_0^q, followed by multiplication by w, then by v into M_n^q. The first map kills z_j. Every such product, and hence the transition, is zero. □

Theorem 2.3. There is a canonical isomorphism

$$H^q(X, F)_I^{\wedge} \xrightarrow{\sim} \varprojlim_n H^q(X, F/I^n F).$$

It is a homeomorphism for the quotient inverse-limit topologies.

Proof. Write $T_n^q = H^q(X, F/I^n F)$. The sheaf short exact sequence gives compatible exact sequences

$$0 \longrightarrow M^q/J_n^q \longrightarrow T_n^q \longrightarrow K_n^{q+1} \longrightarrow 0.$$

For $m \geq n + b_{q+1}$, the transition on the last term is zero. Hence the image of $T_m^q \to T_n^q$ is contained in the first term M^q/J_n^q. It equals that term: transition $M^q/J_m^q \to M^q/J_n^q$ is onto and these modules inject into T_m^q, T_n^q. This proves the Mittag–Leffler assertion and its stable image, with a written uniform bound.

If (u_n) is a compatible family in T_n^q, its image in K_n^{q+1} is zero at every coordinate, because that image is the transition of a sufficiently distant kernel term. Thus u_n has a unique preimage in M^q/J_n^q; uniqueness makes those preimages compatible. Consequently

$$\varprojlim_n T_n^q = \varprojlim_n M^q/J_n^q.$$

Lemma 2.2 gives $I^n M^q \subset J_n^q$ and $J_{n+a_q}^q \subset I^n M^q$. Projection and projection from the shifted index define inverse maps between the two quotient limits. A common finer index proves that their composites are the identity. Each map is continuous coordinate by coordinate, proving the topological assertion. Every construction is the canonical quotient map, so the isomorphism is the asserted one. □

This supplies the critical image, kernel and limit arguments of Stacks 02OA, 02OB and 02OC. In particular, the proof uses the kernel transition bound directly; it does not rely on the incorrect part-number reference presently printed in 02OB. No inverse limit has been interchanged with an unexplained cohomology operation.

Corollary 2.4. If A is complete local Noetherian, every compatible idempotent on the maximal-ideal thickenings of a projective X comes from a unique idempotent of $\Gamma(X, \mathcal{O}_X)$.

Proof. Projective finiteness makes $M = \Gamma(X, \mathcal{O}_X)$ finite. Lemma 1.2 gives $M = M \otimes_A \widehat{A} = \widehat{M}$. Theorem 2.3 identifies this ring with the limit of the thickened section rings. The equality $e^2 = e$ holds in this limit coordinatewise and hence holds for the corresponding actual section. Uniqueness follows from the same ring isomorphism. Idempotents at successive levels are compatible whenever they lift the same initial idempotent, by Lemma 1.4's unique nilpotent lifting. □

3. Smoothness and line bundles at a finite stage

Lemma 3.N (integer finite-stage rings). Every ideal of $\mathbf{Z}$ is principal. Hence $\mathbf{Z}$, its finite-type algebras and their localizations are Noetherian.

Proof. For $d > 0$, induction on a nonnegative integer a gives $a = qd + r$ with $0 \leq r < d$: increment the remainder at each step, replacing d by zero and incrementing the quotient when that remainder reaches d. Negating this equality, and adjusting the quotient and remainder when $r \neq 0$, gives division with such a remainder for every integer a. A nonzero ideal contains a positive integer; let d be its least positive member. The remainder on dividing any member by d belongs to the ideal and is smaller than d, so it is zero. Thus the ideal is $d\mathbf{Z}$; the zero ideal is principal too. The Noetherian module criterion in AG-CA, *Noetherian rings and modules*, Theorem 1.1, now applies to $\mathbf{Z}$. Its Theorem 2.1 proves the assertions for finite-type algebras and localizations. □

Lemma 3.M (finite character presentations). Every finitely generated abelian group is a direct sum of a finite-rank free group and finitely many finite cyclic groups. In particular a finitely generated torsion-free abelian group is free. If M is finitely generated and R is any ring, $R[M]$ is finitely presented and faithfully flat over R. On a nonempty base, finite type of a diagonalizable group conversely forces its exponent group to be finitely generated.

Proof. Choose a surjection $\mathbf{Z}^n \to M$. Its kernel is finitely generated: Lemma 3.N makes $\mathbf{Z}$ Noetherian, and the Noetherian module criterion and submodule proof in AG-CA-03, Theorem 1.1 and Proposition 1.2, apply to $\mathbf{Z}^n$. Write the chosen kernel generators as columns of a finite integer matrix C.

Integer row and column operations diagonalize C. If the matrix is nonzero, take the least positive absolute value of a nonzero entry among all matrices obtainable by invertible integer row and column operations. This least value exists among positive integers. In a matrix attaining it, move such an entry to position $(1, 1)$ and change its sign to make it positive, say d. Every other entry in its first column is divisible by d: otherwise division with remainder and subtraction of a multiple of the first row would give an equivalent matrix with an entry strictly between zero and d. Clear that column by these row operations. The same argument with column operations makes every entry in the first row divisible by d and clears that row. Those column operations leave the cleared column

below d zero. Repeat in the smaller remaining block. Induction gives a diagonal matrix with positive nonzero entries $d_1, \dots, d_s$ and a zero block. Row operations change the basis of $\mathbf{Z}^n$; column operations change the generating list of its kernel without changing that subgroup. Therefore its quotient is $\mathbf{Z}^{n-s} \oplus \bigoplus_{i=1}^{s} \mathbf{Z}/d_i$. Terms with $d_i = 1$ are zero. Every term with $d_i > 1$ has a nonzero torsion element, proving the torsion-free assertion.

For finite presentation one can use the kernel generators directly. If they are $v_1, \dots, v_a \in \mathbf{Z}^n$, then

$$R[M] = R[z_1^{\pm 1}, \dots, z_n^{\pm 1}]/(z^{v_j} - 1)_{j=1}^{a}.$$

Indeed $z^{v+w} - 1 = z^v(z^w - 1) + (z^v - 1)$ and $z^{-v} - 1 = -z^{-v}(z^v - 1)$ put the relation for every kernel vector in this ideal. Distinct cosets give distinct exponent basis elements in $R[M]$. A linear combination mapping to zero has coefficient sum zero within each coset; subtracting a representative monomial in that coset expresses it as a sum of the relations just described. Thus the displayed algebra map has exactly this kernel. Present the Laurent algebra by the $2n$ variables z_i, w_i and the n equations $z_i w_i - 1$; negative powers in the finitely many further equations are represented by the w_i. This is a finite polynomial presentation over R.

The coordinate module $R[M] = \bigoplus_{m \in M} Re^m$ is free. Tensoring an injection acts separately on each exponent summand, proving flatness. Its zero-exponent summand is R, so tensoring a nonzero module remains nonzero, proving faithfulness. Finally, over a nonzero affine base ring, a finite list of algebra generators of $R[M]$ involves only finitely many exponents. Its generated algebra lies in the subgroup algebra generated by those exponents. Independence of the exponent basis forces that subgroup to be all of M. A nonempty base has a nonzero affine chart, so the converse follows there. □

Lemma 3.F (differentials over a perfect field). Let k be perfect, let $t_1, \dots, t_d$ be algebraically independent over k, and let K be a finite extension of $F = k(t_1, \dots, t_d)$. Then $\dim_K \Omega_{K/k} = d$. No separability assumption on K/F is required.

Proof. In characteristic zero, express the finite extension as a finite tower adjoining algebraic elements. An irreducible polynomial has nonzero derivative, and the Euclidean algorithm makes its derivative nonzero at its root. Differentiating that polynomial extends a derivation uniquely across the root: if $f(X) = \sum_i a_i X^i$, its value at a root u must be $-\sum_i \delta(a_i) u^i / f'(u)$. This value also gives a derivation of the polynomial quotient, because it kills its defining relation, and extends to the field by the quotient rule. Iterating the tower proves $\Omega_{K/k} = K \otimes_F \Omega_{F/k}$. Polynomial differentials and localization make the latter module free on $dt_1, \dots, dt_d$.

Suppose the characteristic is $p > 0$. Since k is perfect, $F^p = k(t_1^p, \dots, t_d^p)$. The monomials $\prod_i t_i^{a_i}$ with $0 \le a_i < p$ form a basis of F over F^p: polynomial exponents split uniquely into their multiples of p and their remainders, and localization preserves this basis. Inverting polynomials over the smaller polynomial ring gives all of F, since $h^{-1} = h^{p-1}/h^p$. Hence $[F : F^p] = p^d$. Frobenius is an isomorphism from the extension K/F to K^p/F^p, so $[K^p : F^p] = [K : F]$. Multiplication of finite tower bases, with independence checked by grouping coefficients in the last basis, gives $[K : F^p] = [K : F]p^d = [K : K^p][K^p : F^p]$. Therefore $[K : K^p] = p^d$.

Starting with K^p, adjoin an element u whenever the current field E is smaller than K. Its pth power belongs to E. Its monic minimal polynomial divides $(X - u)^p$ over K, so is $(X - u)^r$ there. If $1 \le r < p$, its coefficient $-ru$ at degree $r - 1$ would put u in E, a contradiction. Thus each nontrivial step has degree p. Exactly d steps produce elements $u_1, \dots, u_d$ with the monomials $\prod_i u_i^{a_i}$, $0 \le a_i < p$, a basis of K/K^p. They give the algebra presentation $K = K^p[U_1, \dots, U_d]/(U_i^p - u_i^p)_{i=1}^{d}$: reduction of powers gives a spanning list of size p^d, and the displayed basis makes it independent. Every k-derivation kills K^p, while $k \subset K^p$. Consequently $\Omega_{K/k} = \Omega_{K/K^p}$. The differential presentation has zero differentials for all its defining relations, so this module is free on $du_1, \dots, du_d$. This proves the formula, including $d = 0$. □

Polynomial calculations for the smooth charts. Over a field k, division by a nonzero polynomial $g \in k[T]$ is proved by repeatedly subtracting the multiple of g that cancels the current

leading term. The degree strictly decreases, so induction gives $f = qg + r$ with $\deg r < \deg g$, or $r = 0$. In a nonzero ideal choose a monic polynomial of least degree; dividing any other member by it makes the remainder a member of smaller degree and hence zero. Thus every ideal of $k[T]$ is principal. A nonzero prime ideal has an irreducible monic generator: a factorization would contradict primality and least degree. Conversely, if g is irreducible and does not divide f, the principal ideal (g, f) has a generator dividing both, which must be a unit; a relation $ag+bf = 1$ then shows that every nonzero class modulo g is invertible. The nonzero prime ideals are consequently maximal, and the localizations at the zero prime and at a nonzero prime have dimensions zero and one respectively.

For $k[T_1, \dots, T_D]$, order monomials lexicographically. The largest monomial in a product is the product of the two largest monomials, with nonzero coefficient equal to the product of their coefficients. Hence this polynomial ring is a domain. Its degree-j monomials correspond to nonnegative integer lists $(a_1, \dots, a_D)$ with sum j. For $D > 0$, inserting $D - 1$ separators among j identical marks counts these lists as $\binom{j+D-1}{D-1}$. Summing from $j = 0$ to $n - 1$, Pascal's identity and induction give $\binom{n+D-1}{D}$. For $D = 0$, only degree zero contributes. These are the polynomial-domain and growth calculations used in the regular-parameter proof.

The cotangent map used there also follows directly from the product rule. Let $P = k[X_1, \dots, X_n]$, let $B = P_{\mathfrak{Q}}$ have maximal ideal $\mathfrak{m}$ and residue field κ, and extend the polynomial derivation by $d(a/u) = u^{-1}da - au^{-2}du$. In $\Omega_{P/k} \otimes_P \kappa$, the differential of a product of two elements of $\mathfrak{m}$ is zero. For $a \in B$ and $f \in \mathfrak{m}$, the same rule gives $d(af) = \bar{a}\, df$. Therefore $[f] \mapsto df$ defines a κ-linear map from $\mathfrak{m}/\mathfrak{m}^2$. An invertible Jacobian minor makes the corresponding differential vectors independent, so their cotangent classes are independent. This argument requires no separability of κ/k, and does not assert that the whole cotangent map is injective.

Flatness on affine charts. A ring map $R \to S$ is flat if every $S_{\mathfrak{q}}$ is flat over $R_{\mathfrak{p}}$, where $\mathfrak{p}$ is the contraction of $\mathfrak{q}$. Indeed, for an injection $U \to V$ of R-modules let K be the kernel after tensoring with S. This is an S-module. Exact localization identifies $K_{\mathfrak{q}}$ with the kernel after tensoring with $S_{\mathfrak{q}}$. That kernel is zero: $R_{\mathfrak{p}}$ is flat over R, and composition with the assumed flat map to $S_{\mathfrak{q}}$ preserves the injection. Local detection makes $K = 0$, proving flatness. Conversely, if S is flat over R, localization and composition make $S_{\mathfrak{q}}$ flat over R. Tensoring $R_{\mathfrak{p}}$-modules with $S_{\mathfrak{q}}$ over R or over $R_{\mathfrak{p}}$ gives the same module, since the inverted elements already act invertibly. Hence $S_{\mathfrak{q}}$ is flat over $R_{\mathfrak{p}}$. This proves both directions over arbitrary rings.

Lemma 3.0 (the split conormal criterion). For a polynomial presentation $C = P/J$, the algebra is formally smooth precisely when $d : J/J^2 \to \Omega_{P/B} \otimes_P C$ has a C-linear left inverse.

Proof. Formal smoothness lifts the identity of C to a section $s : C \to P/J^2$. The map $D(p) = p \bmod J^2 - s(p \bmod J)$ takes values in J/J^2, satisfies the product rule because this ideal is square-zero, and sends $j \in J$ to its class. The universal derivation for the polynomial ring therefore factors it through $\Omega_{P/B} \otimes C$, giving a left inverse of d. Conversely a left inverse gives this derivation by composition with the polynomial universal derivation. The map $p \mapsto p \bmod J^2 - D(p)$ is multiplicative by the product rule and the square-zero condition. It kills J and is the identity after reduction to C, so gives a section. In any square-zero lifting test, lift the polynomial variables arbitrarily. The resulting map from P sends J into the square-zero test ideal and kills J^2; composing its factor through P/J^2 with the section gives the required lift. Polynomial differentials are free on the variable symbols by the product rule, so no unproved splitting assertion enters this argument. □

Lemma 3.A (maps between smooth schemes). Let X and Y be smooth schemes locally of finite presentation over any scheme S, and let $h : X \to Y$ be an S-map. If the map on relative tangent spaces at a point x, with scalars extended to $\kappa(x)$, is surjective, then h is smooth near x. If that map is an isomorphism, then h is étale near x. In particular, on a smooth morphism of relative dimension d, local functions $q_1, \dots, q_d$ whose differentials form a basis at x give an étale coordinate map to $\mathbb{A}^d_S$ near x.

Proof. Work on compatible affine neighbourhoods of x, $h(x)$ and their image in S. Write their algebras as C, D, A. The map $D \to C$ is finitely presented: choose finite presentations of D and C over A, and add, to the presentation of C, the finitely many equations equating each generator of D with a polynomial representing its image in C. This gives a finite presentation over D.

The split conormal criterion makes $\Omega_{C/A}$ and $\Omega_{D/A}$ finite projective. Indeed, in a finite polynomial presentation the conormal injection has a left inverse, so its cokernel is a direct summand of the finite free module of polynomial differentials. Such a summand is flat, and the local finite-freeness proof in AG-RG-S01, Lemma 1.2, applies. Shrink the affine neighbourhoods to trivialize these modules. Dualizing the asserted tangent surjection gives an injection on the fibre of

$$u : C \otimes_D \Omega_{D/A} \longrightarrow \Omega_{C/A}.$$

A full-size minor of its matrix is nonzero at x; invert it. Projecting to the corresponding rows and multiplying by that square matrix's inverse gives a C-linear left inverse of u. In the tangent-isomorphism case its matrix is square and invertible after this shrink.

Consider an arbitrary square-zero surjection $R \to R/J$, an A-algebra map $C \to R/J$, and a prescribed D-algebra structure on R agreeing with it after reduction. Formal smoothness of C/A, as proved by Lemma 3.0, supplies an A-algebra lift $v : C \to R$. Its restriction to D differs from the prescribed map by an A-derivation $\delta : D \to J$: expanding a product leaves the two first-order terms, since $J^2 = 0$. It corresponds to a map $C \otimes_D \Omega_{D/A} \to J$, where the C-action on J is through the given reduced map. Compose it with the left inverse of u to obtain $\Omega_{C/A} \to J$, hence an A-derivation $\widetilde{\delta} : C \to J$. The map $v - \widetilde{\delta}$ is again an A-algebra map by the same square-zero product expansion. Its restriction to D is the prescribed map. Thus C/D is formally smooth. Its finite presentation makes it smooth.

If u is an isomorphism, two such D-lifts differ by a derivation $C \to J$ whose differential map vanishes on the image of u. That image is all of $\Omega_{C/A}$, so the lifts agree. The map is formally étale and finitely presented, hence étale. The coordinate assertion is this case with $Y = \mathbb{A}^d_S$: its pulled-back differentials have basis $dq_1, \ldots, dq_d$, and the given basis condition makes u an isomorphism near x. No Noetherian, reducedness, characteristic or residue-field separability hypothesis was used. □

Lemma 3.1. Let $A = \varinjlim A_i$ be a filtered ring colimit. A finitely presented algebra map $B_0 \to C_0$ over A_0 whose base change over A is smooth becomes smooth after a finite stage.

Proof. Present $C_0 = B_0[x_1, \ldots, x_r]/(f_1, \ldots, f_s)$. At stage i, let $P_i = B_i[x_1, \ldots, x_r]$, $J_i = (f_{1,i}, \ldots, f_{s,i})$ and $C_i = P_i/J_i$. Lemma 3.0, the split conormal criterion of AG-CA-17 Theorem 3.1, says that the map

$$J/J^2 \xrightarrow{d} C^r$$

over the colimit has a C-linear left inverse ρ. Each $\rho(dx_k)$ is a finite sum of classes of the f_j with coefficients in C. Lift these finitely many coefficients to one stage and define $\rho_i : C_i^r \to J_i/J_i^2$ by those images. The equations $\rho_i(df_{j,i}) = f_{j,i} \bmod J_i^2$ are finitely many equations. Their truth in the colimit is witnessed by finite sums of products of the f_j, representing membership in J^2. Lift these witnesses and enlarge the stage until the polynomial equalities hold. Then $\rho_i d$ fixes all generators of J_i/J_i^2, so is its identity. The same criterion gives formal smoothness at that stage, and the existing finite presentation gives smoothness. □

This is the actual finite witness proof in Stacks 0C0B.

Proposition 3.2. A smooth scheme of finite presentation over an affine filtered limit has a smooth finite-presentation model at a finite stage. A projective embedding and a specified line bundle can be retained simultaneously.

Proof. First descend the finite-presentation scheme, the embedding if specified, and finitely many affine source and target charts by AG-MO-03 Theorem 4.1. Retain the closed immersion by its

Theorem 4.2. Choose a finite affine cover of the original source on which the chart algebra maps are smooth. Descend these opens and the maps, and retain their covering property by Lemma 2.1 of that lesson. Apply Lemma 3.1 on the finitely many chart maps and take a common later stage. Smoothness is now true on a cover, hence globally.

For an invertible sheaf, refine the cover so that it is trivial on each chart. On finite affine overlap covers record each transition unit and its inverse. Record their inverse equations, their agreements on overlap refinements, and the cocycle equations on finite covers of triple overlaps. The sections and their finite equality witnesses descend by the section and eventual-equality parts of AG-MO-03. At a common stage these equations hold, so gluing the trivial rank-one sheaves gives an invertible sheaf there. Retain the finite chart covers, smoothness and the projective closed immersion at that same stage. Pullback recovers the original sheaf and scheme. □

This fills the finite-cover step omitted in Stacks 0C0C. It proves only the finite chart and line bundle case being used, without importing general coherent sheaf descent or a general perfection theorem.

4. Geometrically connected smooth affine models

The model needed for the normalizer argument is smooth and affine. We prove this case; the unrestricted finite-presentation connected-fibre theorem is not needed.

Lemma 4.C (Noetherian constructibility and images). In a Noetherian space, a subset is constructible exactly when it is a finite union of locally closed subsets. Equivalently, its intersection with every irreducible closed subset is either not dense or contains a nonempty open of that subset. The image of a constructible subset under a finite type morphism between Noetherian schemes is constructible.

Proof. First every Noetherian space is a finite union of irreducible closed subsets. Otherwise choose, by the descending chain condition, a minimal closed counterexample. It is reducible, hence the union of two proper closed subsets; minimality expresses each as a finite union of irreducible closed subsets, a contradiction. Removing contained members from such a finite list gives its finitely many irreducible components. Every open of a Noetherian space is quasi-compact: among finite unions of members of an open cover choose a maximal one by the ascending chain condition on opens, and maximality forces it to be the whole covered open. Thus all opens are retrocompact. A difference of two opens is locally closed, and an open intersected with a closed subset is such a difference. This proves the first description. On a Noetherian affine scheme, a locally closed subset is a finite union of pieces $V(J) \cap D(g)$: write its open part as a finite union of principal opens and its closed part as $V(J)$, with J finite by Noetherianity.

For the criterion, a dense finite union of locally closed subsets of an irreducible closed Z has one member with closure Z. That member is a nonempty open in Z, proving necessity. For sufficiency, choose a minimal closed F on which the given subset E has nonconstructible intersection. If F is reducible, its finitely many proper irreducible components have constructible intersections by minimality, giving the contradiction on their union. If $E \cap F$ is not dense, use its proper closed closure and minimality. Otherwise the assumed criterion supplies a nonempty open $W \subset F$ lying in E. The rest lies in the proper closed $F \setminus W$ and is constructible by minimality. Together with W it makes $E \cap F$ constructible, the final contradiction. In these steps a subset locally closed in a closed subset is locally closed in the ambient space, so each finite description remains valid there.

We prove the needed generic-image assertion. For domains $A \subset B$ with B of finite type over A, put $K = \operatorname{Frac} A$. The integral domain B_K is nonzero. The complete Noether-normalization proof in AG-CA-09 Corollary 3.2 supplies $K[t_1, \dots, t_d] \subset B_K$ over which B_K is finite. Choose finite algebra generators b_j of B. After inverting a common nonzero $a \in A$, all t_i belong to B_a, and the finitely many monic equations for the b_j have coefficients in $A_a[t_1, \dots, t_d]$ and hold in B_a. Their equalities hold after such a localization because an element zero in B_K is killed by a nonzero base

denominator. Algebraic independence keeps the polynomial subring injective. The monic equations express every monomial in the b_j as a combination of finitely many bounded-exponent monomials, so B_a is finite and integral over this subring, exactly as in AG-CA-05 Theorem 1.1. Its complete lying-over proof, Theorem 3.2, makes its spectrum surject onto that polynomial spectrum. The latter surjects onto $\operatorname{Spec} A_a$, since a polynomial ring over each residue field is nonzero and has a prime. Hence the image contains $D(a)$.

Now let $f : X \to Y$ be finite type between Noetherian schemes. For an irreducible closed $Z \subset Y$ with $f(X) \cap Z$ dense, the closed inverse image has finitely many irreducible components. Their image closures cover Z, so one is dominant. Give it and Z reduced structures. Choose an affine open in Z and a nonempty affine open in its inverse image on that component. This remains a dominant finite type map of integral schemes: their generic points are retained, and restricting to an affine open is quasi-compact in a Noetherian scheme. Its ring map is an inclusion of domains. The generic-image assertion supplies a nonempty open of Z in $f(X)$. The criterion just proved therefore makes the full image constructible. Finally express a constructible subset of X as a finite union of locally closed subsets. Their reduced locally closed structures map to Y by finite type morphisms: a closed immersion is finite type and the open part is quasi-compact on a Noetherian scheme. Apply the full-image result to each and take the finite union. This proves the whole assertion. □

Lemma 4.1. Over an algebraically closed field k, an affine finite type connected scheme remains connected after every field extension. A smooth affine finite type scheme has disjoint irreducible components, and each connected component is integral and remains integral after every field extension of an algebraically closed ground field.

Proof. Connectedness of $\operatorname{Spec} B$ is equivalent to absence of nontrivial idempotents. Indeed an idempotent gives the disjoint opens $D(e), D(1-e)$; conversely a disjoint open-and-closed decomposition gives a section with values one and zero, hence an idempotent by affine sections.

Suppose an extension L/k supplies a nontrivial idempotent $e \in B \otimes_k L$. Express e and $1-e$ using finite lists of k-linearly independent elements of B, with coefficients in L. Let $R \subset L$ be the finite type k-algebra generated by these coefficients, inverted nonzero coefficients ensuring that both expressions stay nonzero, and the finitely many coefficients needed for $e^2 = e$. The relation already holds over R, since $B \otimes_k R \to B \otimes_k L$ is injective. The nonzero finite type algebra R has a k-valued point by the proved Nullstellensatz in AG-CA-06. Specializing at it gives a nontrivial idempotent in B, a contradiction.

For smooth B, every local ring is a regular local domain by AG-CA-18 and AG-CA-14. Two distinct irreducible components cannot meet at a point, since their distinct minimal primes would give distinct minimal primes in that local ring. There are finitely many irreducible components by the Noetherian decomposition proved in Lemma 4.C; they are consequently both closed and open. Smoothness also makes the ring reduced. Each connected component is thus reduced and irreducible, hence integral. After extending an algebraically closed field, connectedness just proved and smoothness preserved by base change give the same conclusion. □

It follows that the number of geometric connected components of a smooth affine finite type scheme is invariant under field extensions. To check this, embed the chosen algebraic closures of the two fields into a common algebraically closed extension. Apply Lemma 4.1 to each of the finitely many open-and-closed components. Extension is faithfully flat and preserves nonemptiness, so each component remains one component.

Lemma 4.A (field and integer calculations before primitive elements). Let F be any field. Polynomial division and Bezout hold in $F[T]$. Minimal-polynomial quotients, distinct-root bounds, the derivative test for a separable algebraic generator, finite splitting fields, and finite tower degree and embedding counts have the following direct proofs. Every finite subgroup of a field's units is cyclic. Every integer ideal is principal and $\mathbf{Z}$ is Noetherian. The integer division, Bezout and prime-factorization facts needed for the group calculation are proved here as well.

Proof. First give the integer arithmetic used in the finite-group calculation. For integers z and $d > 0$, there are q, r with $z = qd + r$ and $0 \le r < d$. For $z \ge 0$, choose the largest q in the finite set $\{0, \dots, z\}$ with $qd \le z$; if the remainder were at least d, the next integer would contradict maximality. For $z < 0$, divide $-z$; negate its quotient when the remainder is zero, and otherwise use quotient $-q-1$ and remainder $d-r$. Thus integer division also covers negative integers. A nonzero subgroup $J \subset \mathbf{Z}$ contains a positive integer and has a least positive member d. The remainder on dividing any member of J by d is in J and smaller than d, so is zero. Hence $J = d\mathbf{Z}$. The zero ideal is generated by zero, so every integer ideal is principal. For an ascending chain of integer ideals, its union is an ideal generated by one integer d. If $d = 0$, every member is zero. Otherwise d lies in one member, which then contains the whole union, so the chain stabilizes. This proves that $\mathbf{Z}$ is Noetherian. For a, b not both zero, apply this to the subgroup of integer combinations $ua + vb$. Its positive generator d divides both a, b, every common divisor divides d, and $d = ua + vb$ for some integers u, v. This proves the positive gcd and Bezout identity. If $\gcd(a, b) = 1$ and $a \mid bc$, multiplication of a Bezout identity by c gives $a \mid c$. A positive prime integer is consequently a prime element. Every integer $n > 1$ factors into positive primes by induction: if it is composite, write it as a product of two smaller positive integers and factor those. Prime divisibility and integer cancellation prove uniqueness. The integer 1 has the empty factorization. The least common multiple of a nonempty finite list of positive integers is therefore the product of the largest occurring powers of each prime; it is 1 if all entries are 1. Different prime powers have gcd one. For coprime positive a, b, the divisibilities $a \mid r$ and $b \mid r$ imply $ab \mid r$: write $r = bc$ and apply the coprime divisibility assertion to obtain $a \mid c$. These are all the integer facts required below.

If $g \neq 0$ and f has degree at least that of g, subtract the leading coefficient of f divided by that of g, times the appropriate power of T times g. This strictly decreases the degree. Induction gives $f = qg + r$ with $r = 0$ or $\deg r < \deg g$. The leading coefficient of a product of two nonzero polynomials is the product of their nonzero leading coefficients, so degrees add. Consequently two such division expressions have the same remainder and quotient. For polynomials a, b not both zero, repeat division on the pair and its successive remainders. Nonnegative degrees strictly decrease, so the process ends. Its last nonzero remainder, scaled to be monic, is d. Back substitution expresses $d = ua + vb$. Each earlier remainder is divisible by d, and any common divisor of a, b divides every remainder, so d is their monic greatest common divisor. In particular coprime polynomials satisfy $ua + vb = 1$. An irreducible f is prime: if f does not divide a, Bezout for f, a, multiplied by b, shows that $f \mid ab$ forces $f \mid b$. Factor existence follows by induction on degree, and prime divisibility gives uniqueness after fixing monic factors and a scalar.

For an algebraic element α, choose a monic nonzero annihilating polynomial of least degree, denoted f. Division shows that every annihilating polynomial is a multiple of f. If f were a product of two nonconstant factors, evaluating in the extension field would make one smaller factor vanish, a contradiction. Thus f is irreducible. Every nonzero class modulo f has a polynomial inverse by Bezout, so $F[T]/(f)$ is a field. Evaluation gives $F[\alpha] = F(\alpha)$ and identifies it with this quotient. The remainders $1, T, \dots, T^{\deg f - 1}$ form a basis: they span by division, and a linear combination of smaller degree cannot be a nonzero multiple of f.

Division by $T - c$ leaves the constant $f(c)$. If c is a root, $f = (T - c)g$. Every other root is a root of g, since its difference from c is a nonzero field element. Induction proves that a nonzero polynomial of degree d has at most d distinct roots, over every extension field. A finite collection of polynomials has a finite splitting field: if one polynomial has a nonsplit nonconstant factor, factor it into irreducibles by the degree induction, adjoin a root of one such factor using the just-proved field quotient, remove the linear factor, and repeat. At most the sum of the original degrees many roots need be adjoined. The extensions at every step are finite by the displayed power basis, and adjoining these roots therefore yields a finite extension in which every original polynomial splits.

The formal derivative obeys the product rule by expanding monomials. In a splitting field, if $f = (T - c)g$, then $f'(c) = g(c)$. Hence c is a repeated root exactly when f and f' both vanish at c. For an irreducible f with $f' \neq 0$, the derivative has smaller degree, so Bezout makes f, f' coprime

and there are no repeated roots. If $f' = 0$ and f is nonconstant, every root is repeated. Thus the minimal polynomial of an algebraic element is separable precisely when its derivative is nonzero, equivalently when $f'(\alpha) \neq 0$.

The square-matrix test used with the differential presentation also follows directly over a field. Define the determinant by its finite signed permutation sum. That sum is linear in each row and each column. With two equal rows its terms cancel in pairs under their transposition, so it is zero; row interchange changes its sign. Consequently adding a multiple of another row leaves it unchanged, and scaling one row multiplies it by the scale. The same assertions hold for columns. Starting with an n by n matrix, move a nonzero entry to the first position by row and column exchanges, scale it to one, and clear the other entries in that row and column by these additions. Repeat on the remaining square block. Every operation is invertible and alters the determinant by a nonzero scalar. The result is a diagonal matrix with some entries one and the others zero. If any zero entry remains, that output coordinate cannot occur in the image, so the map is not surjective. Hence a surjective square map has only ones and has nonzero determinant. Conversely such a diagonal matrix is invertible. The determinant of a triangular matrix is the product of its diagonal entries: expand in the first column for an upper triangular matrix, or in the first row for a lower triangular matrix, and induct. For size zero the determinant is one and the map is surjective.

If $u_1, \ldots, u_r$ is an F-basis of a finite extension E and $v_1, \ldots, v_s$ is an E-basis of L, the $u_i v_j$ span L over F. A relation between them is first a relation between the v_j over E and then, coefficient by coefficient, a relation between the u_i over F. Both sets of coefficients are zero. Thus they form an F-basis and $[L : F] = [L : E][E : F]$. A finite extension has a finite generating list, for example any finite F-basis, and hence a finite simple-generator tower.

To count embeddings, fix a finite generating list $\alpha_1, \ldots, \alpha_m$ of L/F and a finite splitting field H/F of their minimal polynomials over F. Put $E_i = F(\alpha_1, \ldots, \alpha_i)$. The minimal polynomial f_i of α_i over E_{i-1} divides its minimal polynomial over F, by polynomial division. Under any F-embedding $\sigma : E_{i-1} \to H$, the polynomial $\sigma(f_i)$ therefore divides that already split F-polynomial and splits in H. An extension of σ to $E_i = E_{i-1}[T]/(f_i)$ is uniquely specified by the image of α_i, and each distinct root of $\sigma(f_i)$ defines such an extension by evaluation. The evaluation map is injective because its domain is a field. There are at most $[E_i : E_{i-1}]$ such roots, and exactly that many if f_i is separable. An element separable over F stays separable over an intermediate field because its new minimal polynomial divides the old one. Thus a separable extension L/F has exactly $[L : F]$ embeddings into H, by multiplying the tower counts. Conversely, if every step is separable, these counts also give $[L : F]$ embeddings. If some $\beta \in L$ were inseparable over F, its minimal polynomial would have fewer than $[F(\beta) : F]$ distinct roots. Enlarge H to split that polynomial and all polynomials for a generator tower above $F(\beta)$; no new embeddings of L appear, since every generator image was already a root in the original H. Restricting to $F(\beta)$ and applying the upper bound on extensions gives fewer than $[F(\beta) : F][L : F(\beta)] = [L : F]$ embeddings, a contradiction. This proves the transitivity of separability and all finite embedding counts used here. Only a finite splitting field was required.

Let G be any finite subgroup of a field's units. The order of $g \in G$ is the least positive d with $g^d = 1$, which exists by equality of two powers in the finite group. Integer division shows $g^r = 1$ exactly when d divides r. Let e be the least common multiple of the finitely many orders, computed by the maximum exponent of each prime in their factorizations; integer prime factorization and Bezout were proved above. For each prime ℓ dividing e, select an element g_ℓ of order $\ell^s m$ with ℓ not dividing m and ℓ^s the maximal occurring power. Then $h_\ell = g_\ell^m$ has order exactly ℓ^s, by the divisibility test just proved. For commuting elements x, y of coprime orders a, b, if $(xy)^r = 1$, then $x^r = y^{-r}$ belongs to both cyclic subgroups. Its ath and bth powers are one; an integer Bezout identity makes the element one. Thus both a and b divide r, and ab divides r by the same Bezout identity. Since $(xy)^{ab} = 1$, its order is ab. Inducting over the primes shows that $h = \prod h_\ell$ has order e. Every element of G is a root of $T^e - 1$, so the polynomial root bound gives $|G| \leq e$. The e different powers of h give the reverse inequality. Therefore G consists of those powers and is cyclic.

If G is trivial, the empty prime list gives $e = 1$ and $h = 1$, so this case is included. □

Lemma 4.B (finite separable trace and normalization). Let E/F be a finite separable field extension of degree n. Define $\mathrm{Tr}_{E/F}(z)$ to be the trace of multiplication by z on the F-vector space E. It is independent of basis, and the pairing $(z, w) \mapsto \mathrm{Tr}_{E/F}(zw)$ is nondegenerate. If R is a Noetherian normal domain with fraction field F, its integral closure S in E is a finite R-module; for every nonzero $a \in R$, S_a is the integral closure of R_a in E.

Proof. We use the polynomial division, finite splitting-field construction, power bases, tower-degree formula and separable embedding counts proved in Lemma 4.A. The trace calculation itself is supplied here. The free Stacks, Tag 0BIL provides further reading for the nondegenerate pairing.

Choose a finite generating list $\alpha_1, \dots, \alpha_h$ for E/F, put $E_j = F(\alpha_1, \dots, \alpha_j)$, and let f_j be the monic minimal polynomial of α_j over E_{j-1}. Each α_j is separable over F. Choose a finite extension H/F splitting their minimal polynomials $g_j \in F[T]$, by Lemma 4.A. We construct the isomorphism

$$H \otimes_F E_j \longrightarrow \prod_{\sigma : E_j \hookrightarrow H} H, \qquad c \otimes z \longmapsto (c\sigma(z))_\sigma. \tag{T1}$$

For $j = 0$, it is $H \otimes_F F = H$. At the next step the presentation $E_j = E_{j-1}[T]/(f_j)$ identifies $H \otimes_F E_j$ with $(H \otimes_F E_{j-1})[T]/(f_j)$: send T to $1 \otimes \alpha_j$; the powers of T below $\deg f_j$ and the corresponding powers of α_j are bases, so this map is bijective. Applying the previous product decomposition gives the product of $H[T]/(\sigma(f_j))$. All polynomial operations and the inverse product map are componentwise.

Division shows $f_j \mid g_j$ over E_{j-1}, since the remainder would be a polynomial of smaller degree annihilating α_j. Thus $\sigma(f_j)$ divides the already split separable polynomial g_j, and has distinct roots $r_1, \dots, r_m \in H$, with $m = \deg f_j$. Evaluation on these roots gives

$$H[T]/(\sigma(f_j)) \longrightarrow H^m, \quad [v] \longmapsto (v(r_1), \dots, v(r_m)).$$

Its inverse is the explicit interpolation map

$$(b_1, \dots, b_m) \longmapsto \left[\sum_{i=1}^{m} b_i \prod_{\ell \neq i} \frac{T - r_\ell}{r_i - r_\ell} \right]. \tag{T2}$$

Each denominator is nonzero. The displayed polynomial takes the prescribed values, proving one composite is the identity. For the other, reduce a polynomial modulo the monic polynomial to degree below m. Its difference from the interpolation polynomial has m roots, so the root bound proved in Lemma 4.A makes the difference zero. Each resulting component extends σ by sending α_j to the chosen root. Conversely every embedding restricts to a previous component and sends α_j to one of those roots. The tower-degree formula gives exactly $[E_j : F]$ components. This proves (T1), including degree-one steps and the empty tower.

For matrices over a field, direct finite sums give

$$\mathrm{tr}(UV) = \sum_{i,j} U_{ij} V_{ji} = \mathrm{tr}(VU).$$

If P is an invertible change-of-basis matrix, applying this identity to $P^{-1}M$ and P gives $\mathrm{tr}(P^{-1}MP) = \mathrm{tr}(M)$. Thus the trace of multiplication is well defined. Extending scalars leaves the matrix entries and their diagonal sum unchanged. Under (T1) for E, multiplication by z becomes the diagonal map with entries $\sigma(z)$. Hence

$$\mathrm{Tr}_{E/F}(z) = \sum_{\sigma} \sigma(z), \qquad \mathrm{Tr}_{E/F}(zw) = \sum_{\sigma} \sigma(z)\sigma(w). \tag{T3}$$

After extension to H, the pairing is the dot product on H^n. A vector with a nonzero i-th coordinate pairs nontrivially with the i-th coordinate unit vector. The pairing matrix is therefore invertible

over H; its determinant, computed in F, is nonzero, so the matrix is also invertible over F by Lemma 4.A. This proves nondegeneracy. It applies even when the characteristic divides n; no assertion that $\mathrm{Tr}(1) \neq 0$ is used.

We now prove the integral-closure statement, using the closure and transitivity proof in Integral extensions, Theorem 1.2 and the cofactor and Noetherian submodule proofs in S01, Lemmas 1.2 and 1.3. Start with any F-basis of E. If a vector v satisfies $v^s + c_1 v^{s-1} + \cdots + c_s = 0$ with $c_i \in F$, choose a nonzero common denominator $b \in R$. Then bv satisfies a monic equation with coefficients $b^i c_i \in R$. Scaling each basis vector this way gives an integral basis $v_1, \ldots, v_n$. Put

$$G = (\mathrm{Tr}_{E/F}(v_i v_j))_{i,j}, \qquad \delta = \det G \neq 0.$$

Each entry lies in R: every $\sigma(v_i v_j)$ is integral over R, their sum is integral by the earlier closure proof, and that sum lies in F by the definition of trace; normality places it in R. The same argument gives $\mathrm{Tr}_{E/F}(zv_i) \in R$ for $z \in S$. Writing $z = \sum_j a_j v_j$, with $a_j \in F$, gives

$$G(a_j)_j = (\mathrm{Tr}_{E/F}(zv_i))_i.$$

The proved identity $\mathrm{adj}(G)G = \delta I$ implies $\delta a_j \in R$ for every j. Thus S, an R-submodule by the integrality closure proof, is contained in the finite free module $\sum_j \delta^{-1} R v_j$. Noetherianity makes this submodule finite. The basis vectors are in S, so $\mathrm{Frac}\, S = E$. Transitivity also proves normality of S: an element of E integral over S is integral over R and therefore belongs to S.

Finally, elements of S_a are integral over R_a by their monic equations and multiplication by elements of R_a. If z is integral over R_a, write the coefficients of its monic equation as u_i / a^{h_i}, with $u_i \in R$. Choose N with $Ni \geq h_i$ for every coefficient. The monic equation for $a^N z$ has coefficients $a^{Ni - h_i} u_i \in R$. Hence $a^N z \in S$, so $z \in S_a$. This proves the asserted equality of closures. □

Lemma 4.2. Let A be a Noetherian domain with fraction field K. If a smooth affine finite type A-scheme has geometrically integral generic fibre, there is a nonempty principal open of $\mathrm{Spec}\, A$ on which all its fibres are geometrically integral.

Proof. Write the algebra as B. Smoothness makes B A-flat, so $B \hookrightarrow B_K$. Since the latter is a domain, B is a domain. Let $L = \mathrm{Frac}(B_K)$. An étale coordinate chart at the generic point gives algebraically independent $t_1, \ldots, t_d \in L$ with L finite separable over $K(t_1, \ldots, t_d)$: use the standard smooth chart and its zero-relative-dimension coordinate map from AG-CA-17 Theorem 5.1, applied over the field, followed by AG-CA-18 Corollary 6.2 and AG-CA-17 Lemma 7.1. A primitive element u writes

$$L = K(t_1, \ldots, t_d)(u).$$

Here is the complete primitive-element argument, using Lemma 4.A. Let L/F be finite separable of degree n. Its finite generating list $a_1, \ldots, a_r$ has exactly n distinct embeddings $\sigma_1, \ldots, \sigma_n$ into one finite splitting field H. If F is infinite, for each pair $i \neq j$ the form $\sum_t (\sigma_i(a_t) - \sigma_j(a_t)) X_t$ is a nonzero polynomial over H, since the embeddings differ on a generator. Their finite product is nonzero: the leading coefficient of a product of nonzero polynomials over a field is nonzero, iterating by the number of variables. A nonzero polynomial over H does not vanish on all tuples in the infinite subfield F. Indeed, induction on the number of variables first chooses a tuple in F where one nonzero coefficient polynomial is nonzero, and then avoids the finitely many roots in the last variable. Choose $c_t \in F$ where this product does not vanish. The element $a = \sum c_t a_t$ has n distinct images. Its minimal polynomial therefore has degree at least n by the root bound, and at most n because $F(a) \subset L$. The tower-degree equality gives $L = F(a)$. In a nontrivial extension this element cannot be zero; for a trivial extension use 1.

If F is finite, its finite basis makes L a finite set. Lemma 4.A proves that $L^\times$ is cyclic, including the trivial group case. Choose a generator u of this group. Every nonzero element of L is a power of u, and zero belongs to every subfield; hence $F(u) = L$. The generator is nonzero, and for a trivial

extension we may choose 1. Apply this with $F = K(t_1, \dots, t_d)$ and the field $L = \mathrm{Frac}(B_K)$ above to obtain the required u in the displayed simple-extension presentation.

Clear denominators in the minimal polynomial of u, removing the coefficient content, to obtain a primitive polynomial $P \in K[t_1, \dots, t_d, T]$. It is absolutely irreducible. Indeed $B_K \otimes_K \overline{K}$ is a domain, and localizing it gives the domain $L \otimes_K \overline{K}$. The algebra $K[t_1, \dots, t_d, T]/(P)$ embeds in L by its primitive minimal-polynomial presentation. Tensoring this injection with the field $\overline{K}$ keeps it injective. Its target is the just-obtained domain, so the polynomial quotient over $\overline{K}$ is a domain, proving absolute irreducibility without any assertion about contents after field extension.

Here is the needed polynomial Gauss argument. A product of primitive polynomials over a UFD stays primitive: modulo any prime coefficient factor, the two reductions are nonzero polynomials over a domain, so their product is nonzero. Starting with a field and inducting on the number of variables, factor the coefficient content and then factor the primitive part over the fraction field, whose one-variable polynomial ring has Euclidean division. Clear the denominators and normalize each factor to primitive content. The preceding product calculation shows that this gives a factorization over the coefficient ring, and uniqueness follows from the fraction-field uniqueness and the content. Thus each polynomial coefficient ring is a UFD; its primitive minimal polynomial generates the kernel of evaluation in the field extension, as used above.

Enlarge a nonzero localization of A until P has coefficients there. There is a further nonempty localization on which its specialization is absolutely irreducible in every residue field. Here is a finite proof of this assertion. A triangular variable change $x_i = y_i + y_n^{N_i}$ makes the weights of the finitely many monomials distinct, using successive powers of an integer larger than all occurring exponents. The unique largest weight gives a nonzero leading coefficient in y_n. Invert it and divide by it, making P monic of degree $r > 0$ in that variable. Let D_i bound its degrees in the other variables. A nontrivial factorization over a field has monic factors of positive y_n-degrees $e, r - e$, $1 \le e < r$, with other degrees at most D_i; degrees add in each variable over a domain. For each e, introduce the finitely many coefficients of these possible factors and impose the coefficients of $P - Q_1 Q_2$. This gives a finite type A-algebra R_e. Its generic fibre is zero by absolute irreducibility and the Nullstellensatz over $\overline{K}$. Therefore $1 = 0$ after tensoring with K, so some nonzero $a_e \in A$ makes $(R_e)_{a_e} = 0$. Inverting the finite product excludes every nontrivial factorization over every field extension of every residue field. Degree-zero factors in the monic variable are units and are excluded from this list. This is the finite coefficient proof of the free Stacks 0557 result, with only positive factor degrees used.

Put $B' = A[t_1, \dots, t_d, T]/(P)$ over the localization just obtained. Its generic fraction field and that of B are the same. The two generic affine integral schemes have a common nonempty affine principal open: represent the finite generators of each algebra as fractions in the other, invert their finitely many denominators, and invert the denominators needed for these descriptions to be inverse. AG-MO-03's map descent and eventual equality spread this isomorphism after another nonzero base localization. Thus we have $B_h \cong B'_{h'}$, with $h \neq 0$ and nonempty generic open.

Apply AG-FSE Corollary 5.2 in **Flat morphisms** to the finite B-module B/hB, and shrink so that it is A-flat. In the sequence $0 \to B \xrightarrow{h} B \to B/hB \to 0$, flatness of the last term makes multiplication by h injective after every field base change. The smooth fibre is reduced and Noetherian. A nonzerodivisor belongs to no minimal prime: if it did, localization at that minimal prime would make it zero in the field of a reduced Noetherian ring, giving an annihilator outside the prime. Therefore $D(h)$ is dense in every geometric fibre.

The finite type image theorem proved in Lemma 4.C makes the image of the common open constructible. It contains the generic point, so contains a nonempty open of the base. Shrink to a principal open in it; now the common open is nonempty in every fibre, and remains so after field extension. The B'-fibre is geometrically integral by the polynomial argument. Its nonempty open $D(h')$, identified with $D(h)$, is geometrically irreducible. Being dense in the smooth, geometrically reduced B-fibre, it forces that whole fibre to be irreducible and reduced. This proves the lemma. □

Lemma 4.3. For a smooth affine finite type morphism over a Noetherian base, the number of geometric connected components of its fibres is a constructible function.

Proof. By the Noetherian constructibility criterion proved in Lemma 4.C, it suffices to show constancy on a nonempty open of every irreducible reduced closed subscheme of the base. Restrict to such an integral affine base $\operatorname{Spec} A$, with fraction field K.

If the generic fibre is empty, its flat affine algebra injects into zero and is zero, so the component number is constantly zero. Otherwise the generic geometric fibre has finitely many open-and-closed integral components by Lemma 4.1. Their finitely many orthogonal idempotents, summing to one, descend to a finite separable extension K'/K. To justify separability, first express them over $\overline{K}$. In positive characteristic each expression uses finitely many coefficients which become separable after some common p-power. Since an idempotent equals its own p-power, raising its expression to this power puts its coefficients in K^{sep}. In characteristic zero this step is unnecessary. The finite list of algebraic separable coefficients lies in one finite separable extension. Its components are geometrically integral, as checked after extension to $\overline{K}$.

Write $K' = K[v]$ by the primitive element argument in Lemma 4.2. Clear denominators in its monic minimal polynomial and a Bezout identity with its derivative. After a nonzero localization, $A' = A[v]$ is finite free étale over A, with generic fibre the field K'. It is a domain because its monic polynomial presentation is free over A and embeds in K'. The finite generic idempotents extend after inverting finitely many elements of A'; their finite orthogonality and sum equations hold after shrinking, or directly by the generic injection coming from flatness. They split the base-changed smooth algebra into finitely many smooth factors. Each has geometrically integral generic fibre. Apply Lemma 4.2 to all factors and invert a common nonzero $a \in A'$. At every point of $D(a)$ the factors are exactly the geometric connected components.

The determinant of multiplication by a on the finite free A-module A' is nonzero, because this map is invertible on the generic field. Invert this determinant in A. The adjugate makes a invertible on the whole finite étale cover. That cover is surjective, since a nonzero finite free algebra has nonzero fibres, and those fibres have points. Field-extension invariance from Lemma 4.1 now proves the same constant component number on this nonempty base open. This verifies the criterion on every irreducible closed base. Noetherian induction, removing such a nonempty open on each of finitely many irreducible components and then applying the induction hypothesis to their proper closed complement, gives a finite locally closed stratification with constant component count. This proves constructibility as a function, and in particular of its one-component locus. □

This proves the smooth affine case needed from the free chain 055I, 055H, 055G. The hypersurface and flat quotient arguments above supply the generic constancy input explicitly; none of those tags is being substituted for a proof.

Lemma 4.4. Suppose $A = \varinjlim A_i$, A_0 is Noetherian, and $E \subset \operatorname{Spec} A_0$ is constructible. If the image of $\operatorname{Spec} A$ lies in E, the image of $\operatorname{Spec} A_i$ lies in E at some finite stage.

Proof. Lemma 4.C expresses the complement as a finite union of pieces $V(J) \cap D(g)$, with J finitely generated. Its pullback has empty spectrum, so $(A/JA)_g = 0$. This says $g^m \in JA$ for some m, with a finite expression in generators of J. Its coefficients and the finite equality descend to a stage. The same localized quotient there is zero, so that complement piece has empty pullback. Take a common stage for the finitely many pieces. □

This is a direct affine proof of the needed Stacks 05F4 step.

Theorem 4.5. A smooth affine finite-presentation scheme with geometrically connected fibres over an arbitrary affine ring has a smooth affine model with geometrically connected fibres over a finitely generated $\mathbf{Z}$-algebra. Finitely many specified maps, group laws and closed immersions can be retained.

Proof. Descend the finite presentations and finite identities to A_0, a finitely generated $\mathbf{Z}$-subalgebra of the given ring, using AG-MO-03. Retain smoothness by Section 3 and closed immersions by the

earlier proved closed-immersion descent. On this Noetherian model let E be the locus of exactly one geometric connected component. It is constructible by Lemma 4.3. Its pullback contains the image of the original base, by field-extension invariance and the original hypothesis. Apply Lemma 4.4 to enlarge the coefficient ring to a finite stage whose whole image lies in E. Pullback of the model at that stage has geometrically connected fibres and remains smooth. Previously descended finite identities and closed immersions remain true under this base change. □

This is the consumed smooth affine case of Stacks 05FI. It applies simultaneously to the smooth affine group and its smooth affine subgroup in the normalizer lemma. Projectivity of quotient fibres has not been imposed on unrelated fibres of the model.

5. A universal finite complex for a line bundle

The following is the previously written finite complex proof, retained here before its uses.

Lemma 5.1. Let A be Noetherian and C a complex of flat A-modules concentrated in $[a, b]$, with finite cohomology modules. There is a complex K of finite projective modules in $[a, b]$ and a quasi-isomorphism $K \to C$ which remains a quasi-isomorphism after tensoring with every A-module.

Proof. Construct a bounded-above complex P of finite free modules and a map $f : P \to C$, descending from degree b. With terms above j chosen, the cone has terms

$$D^i = C^i \oplus P^{i+1}, \qquad d(c, p) = (d_C c + f(p), -d_P p).$$

Its cohomology is finite: the cone long exact sequence uses finite cohomology of C and finite cohomology of the finite portion of P chosen so far. Choose finitely many cocycles (c_ℓ, p_ℓ) generating $H^j(D)$. Let P^j be free on e_ℓ, and set $f(e_\ell) = c_\ell$, $d_P(e_\ell) = -p_\ell$. The cocycle equations give $d_P^2 = 0$ and $d_C f = f d_P$. The new cone differential sends $(0, e_\ell)$ to (c_ℓ, p_ℓ), killing H^j without changing higher cohomology. Continue to the left. Each degree stabilizes, giving a quasi-isomorphism $P \to C$.

Its cone is acyclic, bounded above and termwise flat. Such a complex stays acyclic after every tensor: start at its last degree and descend through $0 \to Z^i \to D^i \to Z^{i+1} \to 0$. The quotient is flat, so the kernel is flat and the sequence remains exact under tensor. Thus $H^i(P \otimes_A M) = H^i(C \otimes_A M)$ for every M.

Put $Q = \operatorname{coker}(P^{a-1} \to P^a)$. The left tail is a free resolution of Q, and its first positive homology after tensoring with every M is $H^{a-1}(P \otimes_A M) = 0$, because $C \otimes_A M$ starts in degree a. Consequently Q is flat: tensor that free resolution with a short exact sequence $0 \to M' \to M \to M'' \to 0$. Its free terms give a short exact sequence of complexes; the homology exact sequence and the just-proved degree-one vanishing give an injection $Q \otimes M' \to Q \otimes M$. This is the definition of flatness.

The finite presentation of Q makes it locally finite free by Algebra and sheaf cohomology before reductive groups, Lemma 1.2. It is finite projective as follows. Take a surjection $A^r \to Q$ and a finite principal cover on which it splits. A local splitting extends after multiplication by a power of that chart element, since Hom from a finite presented module commutes with localization: use its finite presentation and exact localization to identify the kernel describing Hom. Increase that power until its composite with the surjection is the same power times the identity globally; equality can be checked on finitely many generators. Thus there are maps $\tau_i : Q \to A^r$ whose composites are $f_i^{N_i} \operatorname{id}_Q$. These powers generate the unit ideal, since the $D(f_i)$ cover. A linear combination of the τ_i whose coefficients combine the powers to one is a global splitting. Hence Q is a direct summand of a finite free module and is finite projective. Set $K^a = Q$, $K^i = P^i$ for $a < i \leq b$. The map to C factors through Q since $C^{a-1} = 0$. Truncation preserves cohomology, and the same acyclic flat cone argument proves the assertion after every tensor. □

Proposition 5.2. Let $X \to S$ be smooth projective of finite presentation and L invertible. If all positive cohomology of L on every geometric fibre vanishes, then $f_* L$ is finite locally free, all

positive higher direct images vanish, and these statements commute with every base change.

Proof. Work over $S = \operatorname{Spec} A$. Section 3 gives a smooth projective model $X_0 \to \operatorname{Spec} A_0$, A_0 Noetherian, and an invertible L_0 pulling back to L. A finite affine cover of X_0 has affine intersections. Its section complex C_0 computes cohomology by the earlier finite-cover and affine-vanishing proofs. Its terms are A_0-flat: an invertible module is projective over its chart algebra, and that algebra is base-flat by smoothness. Projective finiteness makes its cohomology finite. After every algebra base change the section complex is $C_0 \otimes_{A_0} B$, by affine modules and localization. Lemma 5.1 therefore gives a bounded finite projective K_0 computing cohomology after every such change.

Put $K = K_0 \otimes_{A_0} A$, and trivialize its terms near a point s. If a differential has an entry invertible at s, shrink and use row and column operations to make it an identity block. The equation $d^2 = 0$ makes the adjacent differentials zero on that block, splitting off a contractible pair $A \xrightarrow{1} A$. After finitely many cancellations the remaining differentials vanish over $\kappa(s)$. Each remaining free term tensored with $\kappa(s)$ is then the cohomology in its degree. In positive degrees it is zero by the hypothesis and faithful field extension; in negative degrees it is zero because the section complex begins in degree zero. All remaining terms other than degree zero have rank zero. The degree-zero term is finite free. Cancelled pairs remain contractible under every tensor, so the same description computes all base changes. It proves the assertions on every affine open and every affine base change. The canonical maps from sections and the section complexes commute with restriction, so glue to the claimed direct-image and base-change isomorphisms. □

Vanishing was required on the original fibres only. No vanishing on unrelated Noetherian model fibres, reduced-base assumption or flat-base-change assumption was used.

6. The smooth curve regular immersion calculation

The following retains the full independently authored CC0 proof of AG-QC-14, Sections 3–5, with its formerly general proper-finiteness input replaced by the projective proof already given in Algebra and sheaf cohomology before reductive groups. Its injective and finite-cover foundations are proved in that preceding support, Sections 3–5. We first record the vector-bundle Ext identity used in the argument. For finite locally free V, tensor-Hom adjunction gives $\mathcal{H}om(V, I) = V^\vee \otimes I$. Tensoring by V is exact, so its right adjoint $V^\vee \otimes -$ takes injectives to injectives: Hom into it is Hom from an exact functor into an injective. Applying this adjunction to an injective resolution gives

$$\operatorname{Ext}^j_P(V, G) = H^j(P, V^\vee \otimes G).$$

Hom into each injective is contravariantly exact. A short exact sequence in its first argument therefore gives a short exact sequence of Hom complexes, whose cohomology gives the connecting maps and long exact sequence used below.

6.0. Ambient projective-space trace and duality

Put $P = \mathbf{P}^n_k$ for a field k, and define

$$\omega_P = \mathcal{O}_P(-n-1).$$

Its identification with $\det \Omega_P$ is explicit. On $U_i = D_+(T_i)$, use the ordered frame $\sigma_i = (-1)^i \bigwedge_{j \neq i} d(T_j/T_i)$. On $U_0 \cap U_i$, the substitution $T_0/T_i = 1/(T_i/T_0)$, $T_j/T_i = (T_j/T_0)/(T_i/T_0)$ gives Jacobian determinant $(-1)^i (T_i/T_0)^{-n-1}$; thus $\sigma_i = (T_i/T_0)^{-n-1}\sigma_0$. These are exactly the transitions of $\mathcal{O}_P(-n-1)$, proving the identification. For $n = 0$, the differential determinant and every twist are both the one-dimensional trivial line on a point. The cohomological normalization uses the ordered standard affine cover $D_+(T_0), \dots, D_+(T_n)$, with alternating Čech differential. For

$n \geq 1$, the projective-space calculation gives

$$H^n(P, \omega_P) = k \cdot \left[\frac{1}{T_0 \cdots T_n}\right].$$

Define the trace $\mathrm{tr} : H^n(P, \omega_P) \to k$ by sending that class to one. For $n = 0$, P is a point; the same expression is the generator of the degree -1 part of $k[T_0, T_0^{-1}]$, so this convention still identifies its one-dimensional space of sections with k.

Multiplication followed by this trace pairs

$$H^0(P, \mathcal{O}(d)) \ \times \ H^n(P, \mathcal{O}(-d-n-1)) \longrightarrow k.$$

For $n \geq 1$ and $d \geq 0$, the first basis consists of T^b, where $b_i \geq 0$ and $\sum b_i = d$. The matching top-cohomology basis consists of $T_0^{-b_0-1} \cdots T_n^{-b_n-1}$. Their products have trace one precisely for matching exponent vectors, and zero otherwise. For $d < 0$, both spaces in the trace pairing vanish. Thus the trace pairing is perfect for every d. When $n = 0$, both spaces are one-dimensional for every twist, and multiplication has the same conclusion. Negative twists on a point do not vanish.

More generally, projective-space cohomology and this coefficient computation give perfect pairings

$$H^j(P, \mathcal{O}(d)) \ \times \ H^{n-j}(P, \mathcal{O}(-d-n-1)) \longrightarrow k$$

for all j: intermediate groups vanish and the two end cases are the computed pairing and its transpose. Maps between direct sums of twists are matrices of homogeneous polynomials. Multiplication is associative, so these pairings commute with all such matrices, including matrices between different twists. This naturality is the part needed to pass from twists to arbitrary coherent sheaves.

Duality in degree zero

For any coherent F, a morphism $u : F \to \omega_P$ induces a functional on its top cohomology:

$$\theta_F^0(u) = \mathrm{tr} \circ H^n(u).$$

This construction is contravariantly natural in F.

Ambient Lemma 6.A. The trace map is an isomorphism

$$\mathrm{Hom}_P(F, \omega_P) \ \cong \ H^n(P, F)^\vee$$

for every coherent F.

Proof. Choose a surjection $P_0 \twoheadrightarrow F$ with P_0 a finite sum of twists, by the proved finite generation by twists in Algebra and sheaf cohomology before reductive groups, Proposition 8.2. Its coherent kernel K is likewise a quotient of a finite sum of twists P_1. Thus $P_1 \to P_0 \to F \to 0$ is a presentation. Hom into ω_P gives

$$0 \to \mathrm{Hom}(F, \omega_P) \to \mathrm{Hom}(P_0, \omega_P) \to \mathrm{Hom}(P_1, \omega_P).$$

The standard cover has $n+1$ affine opens with affine intersections. Consequently cohomology of quasi-coherent sheaves vanishes above n. The two short exact sequences defining this presentation therefore give

$$H^n(P, P_1) \to H^n(P, P_0) \to H^n(P, F) \to 0.$$

Dualizing gives another left-exact sequence in the same direction as the Hom sequence. All spaces are finite-dimensional, by the proved projective finiteness in Algebra and sheaf cohomology before reductive groups, Theorem 8.4. The maps θ^0 give a morphism of these sequences. The two maps for P_0, P_1 are isomorphisms by the trace pairing, and commute with the presentation map by its polynomial-matrix description. Their kernels are therefore isomorphic, which proves the assertion. □

Only two stages of a presentation were used. In particular, this argument does not claim that repeated coherent kernels eventually become sums of line bundles.

Serre duality in every degree

Ambient Theorem 6.B (Serre duality on projective space). For every coherent sheaf F and every integer i, there are natural isomorphisms

$$\operatorname{Ext}^i_P(F, \omega_P) \;\cong\; H^{n-i}(P, F)^\vee.$$

They extend the trace map, using its fixed trace.

Proof. For $i \geq 0$, consider the contravariant cohomological sequences

$$A^i(F) = \operatorname{Ext}^i_P(F, \omega_P), \qquad B^i(F) = H^{n-i}(P, F)^\vee.$$

Exactness of vector-space duality and vanishing above n make the second a contravariant delta functor starting in degree zero, just as Ext is. Choose $q > 0$ sufficiently large so that $F(q)$ is generated by its sections and all positive cohomology of $\omega_P(q)$ vanishes. There is an epimorphism

$$Q = \mathcal{O}_P(-q)^{\oplus m} \twoheadrightarrow F$$

with coherent kernel K. The vector-bundle identity above and projective Serre vanishing in Algebra and sheaf cohomology before reductive groups, Theorem 8.4 give $A^j(Q) = 0$ for every $j > 0$. The twist calculation gives $B^j(Q) = 0$ for every $j > 0$: if $0 \leq n - j < n$, the corresponding cohomology of a strictly negative twist vanishes, and if $n - j < 0$ it vanishes by convention. This also covers $n = 0$, where every positive B^j is zero.

Both long exact sequences consequently identify their degree-one value on F with the cokernel of their degree-zero map from Q to K. Ambient Lemma 6.A identifies these cokernels. For $i \geq 2$, the sequences give

$$A^i(F) \cong A^{i-1}(K), \qquad B^i(F) \cong B^{i-1}(K).$$

Induction gives the desired isomorphisms for every $i \geq 0$.

These isomorphisms are canonical. The epimorphisms just used efface all positive values of both delta functors. The following comparison gives uniqueness: a morphism in degree zero determines the cokernel map in degree one and then every dimension-shifted map. For two choices of epimorphism, take their fiber product over F, which is coherent, and an epimorphism from sufficiently negative twists onto that fiber product. The resulting maps of short exact sequences compare both choices. The same construction over a morphism of coherent sheaves proves naturality, and compatibility with connecting maps follows from the exact sequences. Thus the induction is the unique extension of the trace map.

If $i < 0$, the left side vanishes, and the right side is cohomology above dimension n, hence also zero. This proves the displayed duality for all integers. □

The following direct DVR argument supplies this local algebra before the degree calculations.

DVR Lemma 6.C. A one-dimensional Noetherian normal local domain is a discrete valuation ring.

Proof. Let $(R, \mathfrak{m})$ be such a ring and choose $0 \neq a \in \mathfrak{m}$. Every prime containing a is $\mathfrak{m}$, since any other nonzero prime would give a chain of length at least two. Thus the radical of (a) is $\mathfrak{m}$. Finite generation of $\mathfrak{m}$ gives $\mathfrak{m}^n \subset (a)$ for some n: choose a power of each generator lying in (a) and use the pigeonhole principle on monomials. Choose the least n and $b \in \mathfrak{m}^{n-1} \setminus (a)$. Then $c = b/a \notin R$ and $c\mathfrak{m} \subset R$.

If $c\mathfrak{m} \subset \mathfrak{m}$, apply the determinant trick to multiplication by c on the finite module $\mathfrak{m}$. Its resulting monic polynomial annihilates this module, hence is zero in $\operatorname{Frac}(R)$ because the domain has a nonzero element in $\mathfrak{m}$. Normality would imply $c \in R$, a contradiction. Consequently cx is a unit for some $x \in \mathfrak{m}$. It follows that $\pi = c^{-1} = x/(cx) \in \mathfrak{m}$ and $\mathfrak{m} = \pi R$.

The intersection $N = \bigcap_j \pi^j R$ is zero. In fact N is a finite ideal, since R is Noetherian, and cancellation of $\pi \neq 0$ shows $N = \pi N$: if $x = \pi y$ lies in every $\pi^{j+1}R$, then y lies in every $\pi^j R$.

Nakayama gives $N = 0$. Repeated division of a nonzero nonunit by π must therefore stop. Every nonzero element is uniquely a unit times π^j, and every nonzero ideal is generated by the element of smallest such exponent. This is the discrete valuation ring assertion. □

DVR Lemma 6.D (the regular curve case). A one-dimensional Noetherian local domain whose maximal ideal is generated by one element is a normal discrete valuation ring. In particular this applies to a one-dimensional regular local domain.

Proof. Write the maximal ideal as πR. The element π is nonzero, since the domain has dimension one and is not a field. The ideal $N = \bigcap_{j\geq 0} \pi^j R$ is finite by Noetherianity. Cancellation shows $N = \pi N$: if $x = \pi y$ lies in every $\pi^{j+1}R$, then y lies in every $\pi^j R$. Nakayama, proved in S01 Lemma 1.2, gives $N = 0$. Repeated division of a nonzero element by π must therefore terminate, giving a unit times π^r for a unique $r \geq 0$. Uniqueness follows because a positive power of π cannot be a unit. A nonzero ideal contains an element with smallest exponent; that element divides every element of the ideal and generates it. This proves the DVR description without assuming normality.

Every nonzero fraction is now uniquely a unit times π^r for an integer r. Suppose a fraction z is integral over R and has negative exponent. In a monic equation $z^m + c_1 z^{m-1} + \cdots + c_m = 0$, divide by z^m. Since $z^{-1} \in \pi R$, every term $c_i z^{-i}$ lies in the maximal ideal, forcing $1 \in \pi R$, a contradiction. Thus the exponent is nonnegative and $z \in R$; the zero fraction is already in R. This proves normality. A one-dimensional regular local domain has a single generator of its maximal ideal by the definition of regularity, so the asserted special case follows. □

Koszul Lemma 6.E. Let $f_1, \ldots, f_c$ be a regular sequence in a ring A, and put $I = (f_1, \ldots, f_c)$. Its Koszul complex is a finite free resolution of A/I, the classes of the f_i form a basis of I/I^2, and for an invertible A-module L its dual complex has cohomology only in degree c, where it is

$$(A/I) \otimes_A L \otimes_{A/I} \det(I/I^2)^*.$$

This last identification agrees under invertible changes of the regular generating list.

Proof. Define $K_r = \bigwedge^r A^c$, with basis e_i and differential

$$d(e_{i_1} \wedge \cdots \wedge e_{i_r}) = \sum_{j=1}^{r} (-1)^{j-1} f_{i_j} e_{i_1} \wedge \cdots \widehat{e_{i_j}} \cdots \wedge e_{i_r}.$$

Pairing the two possible orders of deletion gives $d^2 = 0$. On adjoining f_c, this complex is the mapping cone of multiplication by f_c on the preceding Koszul complex. The cone's short exact sequence gives its homology exact sequence directly by lifting cycles. Induction, starting with the empty list and $K_0 = A$, leaves only degree-zero homology: multiplication by f_c is injective on $A/(f_1, \ldots, f_{c-1})$ by regularity, and its cokernel is A/I. Thus K is the stated resolution.

The map $(A/I)^c \to I/I^2$ is surjective. If $\sum_i a_i f_i \in I^2$, subtract expressions $\sum_j b_{ij} f_j$ from the a_i to make their resulting linear combination of the f_i zero. Exactness at K_1 makes this relation a boundary from K_2; the displayed differential makes every coefficient of such a boundary belong to I. Hence the original a_i also belong to I, proving injectivity and the claimed basis.

The perfect exterior-product pairing identifies

$$\operatorname{Hom}_A(K_r, L) = L \otimes_A K_{c-r} \otimes_A (\bigwedge^c A^c)^*.$$

Define the dual differential by precomposition with d. For $u \in K_{r+1}$ and $v \in K_{c-r}$, the identity $0 = d(u \wedge v) = du \wedge v + (-1)^{r+1} u \wedge dv$ gives $du \wedge v = (-1)^r u \wedge dv$. Thus the dual differential corresponds to $(-1)^r d$ under the displayed pairing; multiplying the degree-r identification by $(-1)^{r(r-1)/2}$ gives the reversed Koszul differential without those signs. Its cohomology therefore vanishes except in degree c, where the canonical top-term quotient is reduction modulo I. The conormal basis just proved identifies that quotient with the displayed determinant line. An invertible matrix changing the f_i gives an isomorphism of the two Koszul complexes by the same basis change

on their exterior terms. The top exterior term changes by its determinant and the dual by its inverse. These are exactly the transition maps of $\det(I/I^2)^*$, so the identification is independent of the chosen regular list. The empty list has determinant A and gives the asserted degree-zero statement as well. □

Lemma 6.1. A closed immersion $i : C \hookrightarrow \mathbf{P}^N_k$ of a smooth pure one-dimensional projective curve is locally cut out by a regular sequence of length $c = N - 1$, and

$$0 \to I/I^2 \to i^*\Omega_{\mathbf{P}^N/k} \to \Omega_{C/k} \to 0$$

is exact.

Proof. First extend k to an algebraic closure. At a closed point the two local rings are regular, of dimensions N and one, by the proved smooth field criterion. AG-CA-14 Proposition 1.4 gives a regular sequence of c parameters generating the ideal. Their linear classes are precisely the kernel of the surjection of tangent cotangent spaces. Thus their differentials are independent in the ambient differential fibre. The complete relation calculation in Koszul Lemma 6.E makes the conormal module free on their classes. The conormal map therefore has rank c on a neighbourhood of that closed point. Its cokernel is $\Omega_{C/k}$ by the proved differential conormal sequence. Since these finite type neighbourhoods cover every closed point, they cover the curve; every nonempty closed complement in a finite type scheme over a field has a closed point by the Nullstellensatz.

Here is the descent back to k. The finite module I/I^2 becomes locally free of rank c after this faithful field extension, so it is flat before extension: any kernel of a tensor injection vanishes after the faithfully flat field extension, hence vanishes already. A finite presented flat module is projective by the earlier module theorem. Locally choose a basis and lift it to c generators of I. At each ambient local ring, $I = (f_1, \dots, f_c) + I^2$; Nakayama applied to the finite module $I/(f_1, \dots, f_c)$, with I in the maximal ideal, gives generation of I. The corresponding Koszul complex becomes exact in positive degrees after the field extension, because any minimal c-generator list differs by an invertible matrix from the parameter list already obtained. Its finite cohomology modules therefore vanish before extension by faithful flatness. Equivalently the list is regular: the Koszul cone for the last generator makes multiplication by that generator surjective on every positive homology module of the preceding Koszul complex. Those modules are finite and the generator lies in the maximal ideal, so Nakayama kills them. The same exact sequence gives injectivity of the last generator on degree-zero homology. Induction proves regularity. Finally exactness of the conormal sequence descends by faithful flatness as well. □

This is the field-curve case actually needed from the more general free Stacks 067U regular immersion theorem. It uses the existing regular quotient proof, rather than importing the full syntomic arbitrary-base theorem.

Theorem 6.2. For such a curve and an invertible L, there is a natural isomorphism

$$H^1(C, L)^\vee \cong H^0(C, \Omega_{C/k} \otimes L^{-1}).$$

Proof. Lemma 6.1 gives the local regular sequence, and the complete resolution, dual and change-of-generators calculation in Koszul Lemma 6.E gives its Koszul resolution. Dualizing into $\omega_{\mathbf{P}^N} = \det \Omega_{\mathbf{P}^N/k}$ has cohomology only in degree c, with that sheaf equal to

$$i_*\big(L^{-1} \otimes i^*\omega_{\mathbf{P}^N} \otimes \det(I/I^2)^*\big).$$

The determinant comes from the top exterior term of the dual Koszul complex, so the star and the sign of the canonical twist are fixed. Taking determinants of the conormal exact sequence identifies $i^*\omega_{\mathbf{P}^N} \otimes \det(I/I^2)^*$ with $\Omega_{C/k}$. Here is the full consumed local-to-global comparison from AG-QC-14 Proposition 2.1, with its finite filtered-complex input supplied by Algebra and sheaf cohomology before reductive groups, Section 4. Resolve $\omega_{\mathbf{P}^N}$ by injectives $I^\bullet$ and put $E^\bullet = \mathcal{H}om(i_*L, I^\bullet)$. Each

term is flasque: on $U \subset V$, extension by zero gives a stalkwise injection $j_!((i_*L)|_U) \hookrightarrow (i_*L)|_V$, and injectivity of $I^q|_V$ extends every morphism from that submodule. Thus Hom sections restrict surjectively. Form the double complex of sections of E^q on the finite standard affine intersections of projective space. Its rows have Čech cohomology only in degree zero, since the terms and all their open restrictions are flasque and the earlier finite-cover comparison applies. One filtration therefore identifies total cohomology with $H^*\Gamma(E^\bullet) = \operatorname{Ext}^*_{\mathbf{P}^N}(i_*L, \omega_{\mathbf{P}^N})$.

On any affine intersection, choose a resolution of i_*L by finite free sheaves, allowing an infinite left tail; Noetherianity makes its successive kernels finite. The first-quadrant Hom double complex with $I^\bullet$ compares its Hom into ω with $E^\bullet$. In the resolution direction injectivity makes Hom exact. In the other direction its finite free source reduces to finite sums of ω, whose positive cohomology on this affine is zero by affine vanishing. Hence the vertical cohomology of the section double complex is the section module of the locally calculated sheaf Ext. Lemma 6.1 and the dual Koszul calculation above show that it is zero except in degree c. The other filtration therefore has only that row, the finite Čech complex of $i_*(L^{-1} \otimes \Omega_{C/k})$. Each total degree has finitely many terms, so the finite filtration calculation gives convergence and no completion or infinite-product issue. In total degree c it gives

$$\operatorname{Ext}^c_{\mathbf{P}^N}(i_*L, \omega_{\mathbf{P}^N}) = H^0(C, L^{-1} \otimes \Omega_{C/k}).$$

Ambient Theorem 6.B, the full trace and dimension-shifting proof of AG-QC-14 Theorem 5.1 reproduced above, identifies the same group with $H^{N-c}(\mathbf{P}^N, i_*L)^\vee = H^1(C, L)^\vee$. The Koszul determinant, conormal determinant and trace are natural, giving the stated isomorphism. □

Lemma 6.3 (divisors, Euler characteristic and genus). Suppose now that k is algebraically closed and C is connected. Put $g = \dim_k H^1(C, \mathcal{O}_C)$. For every invertible L,

$$\chi(C, L) = \deg L + 1 - g, \qquad \deg \Omega_{C/k} = 2g - 2.$$

A line bundle of negative degree has no nonzero section.

Proof. Smooth local rings are regular domains. Thus different irreducible components cannot meet, and the finitely many components are open and closed. Connectedness makes C integral. Projective finiteness makes $H^0(C, \mathcal{O}_C)$ a finite-dimensional domain over k. It is a field, since multiplication by a nonzero element is an injective endomorphism of a finite-dimensional vector space and hence is surjective. Algebraic closedness makes this field k.

There are two affine opens covering C. In a projective embedding choose a hyperplane not containing C; its intersection with the curve is finite, since a proper closed subset of a Noetherian integral curve has dimension zero and finitely many points. For this finiteness, every Noetherian closed subset is a finite union of irreducible closed subsets: a minimal counterexample splits into two proper closed subsets and contradicts minimality. In dimension zero those irreducible subsets are single closed points. Choose another hyperplane avoiding the finitely many intersection points. The field k is infinite: for a finite field the polynomial $\prod_{a \in k}(T - a) + 1$ would have no root. Each forbidden hyperplane condition is a nonzero linear equation on its coefficients; their product is a nonzero polynomial. Induction on the number of variables, using the polynomial root bound in the last variable, gives coefficients where that product is nonzero. The complements of these hyperplanes on C are affine and cover C. Their intersection is affine by separatedness. The proved finite Čech computation consequently gives $H^q(C, L) = 0$ for $q > 1$. All the remaining groups are finite by projective finiteness, so their Euler characteristic is defined and additive in short exact sequences.

A nonzero rational section of L exists by choosing a generator at the generic point. Each closed-point local ring is a DVR: a one-dimensional regular local ring has its maximal ideal generated by a parameter, and Lemma 6.D proves directly that this domain is a normal DVR, without a prior normality assumption. Relative to a local generator, the section has an integer valuation. There are

only finitely many nonzero valuations, since a finite trivializing affine cover represents it by rational functions with finitely many zeros and poles. These valuations define a divisor $D = \sum_x n_x x$ and identify L with $\mathcal{O}_C(D)$: at x, the identification sends the local module $\pi_x^{-n_x}\mathcal{O}_{C,x}$ to the section's generated fractional module. The descriptions agree at the generic point and hence on overlaps.

For every divisor and closed point there is an exact sequence

$$0 \longrightarrow \mathcal{O}_C(D) \longrightarrow \mathcal{O}_C(D+x) \longrightarrow k_x \longrightarrow 0.$$

At x the quotient is $\pi^{-n_x-1}R/\pi^{-n_x}R \cong k$; elsewhere it is zero. The skyscraper k_x has $H^0 = k$ and no higher cohomology (its restrictions are surjective, so the previously proved flasque acyclicity applies). Additivity, successively adding or subtracting points, gives

$$\chi(C, \mathcal{O}_C(D)) = \sum_x n_x + \chi(C, \mathcal{O}_C).$$

This both proves that the integer $\deg L = \sum n_x$ is independent of the rational section and gives the first formula. Tensor product adds these divisors, so degrees add and inversion negates degree. A nonzero regular section has only nonnegative valuations, which proves the negative-degree assertion.

Theorem 6.2 with $L = \mathcal{O}_C$ gives $h^0(\Omega_{C/k}) = g$. The same theorem with $L = \Omega_{C/k}$ gives $h^1(\Omega_{C/k}) = h^0(\mathcal{O}_C) = 1$. Its Euler characteristic is therefore $g - 1$. The first formula applied to $\Omega_{C/k}$ gives $g - 1 = \deg \Omega_{C/k} + 1 - g$, as claimed. In particular $T_C = \Omega_{C/k}^{-1}$ has degree $2 - 2g$. □

Lemma 6.4 (degree under a separable curve map). Let $f : D \to C$ be a nonconstant map between smooth connected projective curves over algebraically closed k, and suppose the function-field extension is separable of degree n. Then f is finite and locally free of rank n, and $\deg f^*L = n \deg L$.

Proof. Put $F = k(C)$ and $E = k(D)$. The hypothesis gives the finite separable extension E/F of degree n. By the smooth connected curve argument in Lemma 6.3, both curves are integral and their local rings are normal. For each nonempty affine open $U = \operatorname{Spec} R \subset C$, R is a Noetherian normal domain with fraction field F. Here is the global normality check: a fraction integral over R belongs to every maximal localization by its already proved DVR normality; a proper denominator ideal for that fraction would lie in a maximal ideal, contradicting the denominator outside that ideal supplied by membership in the localization. Let S be its integral closure in E. Lemma 4.B proves that S is finite over R, normal, and has fraction field E, and that these closures agree on principal overlaps. They therefore glue to a finite normal integral curve $q : C' \to C$.

Here are the local properties needed for this construction. A finite R-algebra is Noetherian: each of its ideals is an R-submodule of a finite R-module, so is finitely generated over R, and those same generators generate it as an ideal. The integral-extension lying-over and incomparability proofs in Integral extensions, Theorems 3.2 and 3.3 show that C' has dimension one, and that each nongeneric point is closed with one-dimensional local ring. Indeed, strict prime chains contract to strict chains in the one-dimensional base, while a maximal ideal over a nonzero base maximal ideal is nonzero. A maximal ideal upstairs contracts to a maximal ideal downstairs. Its local ring is therefore a one-dimensional Noetherian normal domain, and is a DVR by Lemma 6.C. The generic local ring is E.

The given map factors through C'. For $d \in f^{-1}(U)$, the inclusion $R \to \mathcal{O}_{D,d}$ sends any monic equation of $s \in S$ to a monic equation in this normal local domain. As $s \in E$, it follows that $s \in \mathcal{O}_{D,d}$. Such a rational function is regular on a neighbourhood of d: its local fraction presentation has an invertible denominator after shrinking. Thus each s is a section on $f^{-1}(U)$. These functions define the factorization $g : D \to C'$, and the localization equality in Lemma 4.B makes the factorizations agree on overlaps. On function fields g induces the identity of E.

There is an inverse rational map $C' \dashrightarrow D$. Choose a nonempty affine open of D. Its finite

algebra generating list consists of elements of E; on an affine open of C', clear their finitely many denominators. This defines a morphism on a nonempty open with the inverse function-field map. Embed D in a projective space and represent its generic point by finitely many homogeneous coordinates in E. At a nongeneric point of C', its DVR valuation has a least value among the nonzero coordinates. Multiply all coordinates by the inverse of that parameter power. They then belong to the local ring and one is a unit. They define a map there, and their local fraction presentations spread to a neighbourhood with that coordinate invertible. Every homogeneous equation defining D still vanishes, since it vanishes in E and the neighbourhood is integral. The rational map consequently extends to a map into D at every point.

These local extensions glue, and the two composites are identities. To justify both assertions, two maps from an integral scheme to a separated scheme that agree generically agree everywhere: their equalizer is the inverse image of the closed diagonal. On each affine open of the source, every equation of this equalizer is zero in the function field and therefore zero in the domain. For gluing use that D is separated. For the composites use that both D and C' are separated; the latter is finite over the separated curve C, hence separated. This last assertion also follows on affine pieces because the diagonal of an affine morphism is induced by the surjective multiplication map $S \otimes_R S \to S$. Thus g is an isomorphism and f is finite.

The finite algebra of f is torsion-free over each local domain of C, since it embeds in E. A finite torsion-free module M over a DVR is free: lift a basis of $M/\pi M$, and Nakayama gives generators. If there were a nonzero relation, divide all its coefficients by the smallest occurring parameter power. Torsion-freeness makes the divided relation valid, while reduction modulo π gives a nonzero relation in that residue-field basis, a contradiction. At the generic point the module is the F-vector space E, so every such free rank is n. These bases hold on neighbourhoods: the cokernel of a proposed basis map is finite and vanishes at the point, hence vanishes after shrinking; its kernel is finite by Noetherianity and vanishes there as well. Clearing denominators of generators kills both on a common neighbourhood. Hence f is locally free of rank n.

Let x be a closed point of C. Its fibre algebra A has dimension n over k. Every prime quotient of A is a finite-dimensional domain over k, hence a field by the injective-multiplication argument of Lemma 4.A. Each residue field is k: every element is algebraic, and its irreducible minimal polynomial over the algebraically closed k has degree one. There are finitely many maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_r$. Indeed, distinct maximal ideals are comaximal, and any list of length t gives a surjection $A \to k^t$ by the Chinese remainder calculation, forcing $t \le n$. For completeness, if $I + J = A$, choose $u \in I, v \in J$ with $u + v = 1$. The pair of residues represented by a, b has lift $va + ub$; the kernel is $I \cap J = IJ$, because $z = z(u + v)$ for $z \in I \cap J$. Iterating proves the asserted surjection.

The nilradical $N = \bigcap_i \mathfrak{m}_i$ is nilpotent. It has finitely many generators, as A is Noetherian, and each generator is nilpotent; a sufficiently long monomial in the generators must contain a vanishing power. Choose q with $N^q = 0$. The ideals $\mathfrak{m}_i^q$ remain pairwise comaximal: expand $1 = (u + v)^{2q-1}$ with $u \in \mathfrak{m}_i, v \in \mathfrak{m}_j$, and each term belongs to one of those two powers. Their intersection equals their product, which is $(\prod_i \mathfrak{m}_i)^q = N^q = 0$. The same Chinese remainder calculation therefore decomposes A into its local factors, equivalently its localizations at the fibre points.

At a point y over x, let π_x be a parameter of $\mathcal{O}_{C,x}$. In the source DVR its image is a unit times $\pi_y^{e_y}$ for an integer $e_y > 0$: it is nonzero by the function-field injection and belongs to the maximal ideal by locality. The corresponding fibre factor is $\mathcal{O}_{D,y}/(\pi_x)$, whose filtration by powers of π_y has exactly e_y successive quotients equal to its residue field k. Its dimension is therefore e_y, and the decomposition gives $\sum_{f(y)=x} e_y = n$. Pulling back the local equation of the divisor x gives $\sum e_y y$. Lemma 6.3 represents every invertible sheaf L by a divisor. Apply this calculation to each of its finitely many coefficients, including negative coefficients, to obtain $\deg f^* L = n \deg L$. $\square$

Proposition 6.5. A smooth connected projective curve over algebraically closed k dominated by $\mathbf{P}^1_k$ has genus zero. A genus-zero smooth connected projective curve over such k is isomorphic to $\mathbf{P}^1_k$.

Proof. First suppose $f : \mathbf{P}^1 \to C$ is separable of degree n. The differential map $f^*\Omega_{C/k} \to \Omega_{\mathbf{P}^1/k}$ is generically an isomorphism. To prove the field calculation directly, a k-derivation on the smaller function field extends uniquely across a separable generator u with minimal polynomial $g(T) = \sum_i a_i T^i$: its value on u must be $-\sum_i \delta(a_i)u^i/g'(u)$. The denominator is nonzero by S02 Lemma 4.A; that value kills the polynomial relation and therefore defines a derivation of the quotient and of its fraction field. Iterating a finite generator tower proves that the absolute differential module upstairs is scalar extension of the one downstairs, by the universal differential property. Both have rank one by S02 Lemma 3.F. A generically nonzero map of line bundles on an integral smooth curve is injective and has an effective zero divisor R. Its local matrix is a nonzero DVR element, whose valuation is the coefficient of that divisor. Thus

$$-2 = \deg \Omega_{\mathbf{P}^1/k} = n(2g-2) + \deg R.$$

Here the degree on $\mathbf{P}^1$ is -2, either by the two-coordinate calculation $dz = -t^{-2}dt$, or by Lemma 6.3 and $H^1(\mathbf{P}^1, \mathcal{O}) = 0$ from the complete twist calculation in S01 Theorem 7.2. Since $n > 0$ and R is effective, $g \geq 1$ is impossible.

In characteristic $p > 0$, an inseparable dominant map also yields a separable dominant map from a projective line. Regard $k(C)$ as a subfield of $k(t)$. Choose the largest e such that $k(C) \subset k(t^{p^e})$. Such an e is bounded: for one nonconstant rational function of $k(C)$, containment in $k(t^{p^e})$ makes its reduced numerator and denominator polynomials in t^{p^e}, so p^e is bounded by their maximum degree. Indeed a coprime presentation in that variable remains coprime after substitution by its polynomial Bezout identity; uniqueness of a reduced fraction then gives the assertion. Put $z = t^{p^e}$. The kernel of d/dz on $k(z)$ is exactly $k(z^p)$. Indeed the basis $1, z, \ldots, z^{p-1}$ over $k(z^p)$ makes the derivative's kernel its constant summand; independence follows by clearing the polynomial denominators in a relation: the summands then have exponents in distinct congruence classes modulo p, so every coefficient is zero. The same reduction writes every polynomial as such a sum, and $h^{-1} = h^{p-1}/h^p$ for any nonzero polynomial h shows that localizing at polynomials preserves this basis, since k is perfect. Thus these elements are a field basis. Maximality supplies $a \in k(C)$ with derivative nonzero. If $a = P(z)/Q(z)$ in coprime form, the polynomial $P(T) - aQ(T)$ has nonzero derivative at z; the minimal polynomial of z over $k(a)$ therefore has nonzero derivative and the extension is separable. The intermediate extension $k(z)/k(C)$ is separable as well, since an element separable over a smaller field remains separable over an intermediate field. Its field embedding gives a rational map $\mathbf{P}^1_z \dashrightarrow C$, extended everywhere by the homogeneous-coordinate DVR argument above. The separable calculation applies and gives genus zero.

Conversely let $g = 0$ and fix $q \in C(k)$. For $m \geq 0$, Lemma 6.3 and duality give

$$h^0(\mathcal{O}_C(mq)) = m + 1,$$

because $\Omega_{C/k}(-mq)$ has degree $-2 - m < 0$. Choose x complementing the constants in $H^0(\mathcal{O}_C(q))$. It is a rational function with its only pole at q, of exact order one. The functions $1, x, \ldots, x^m$ have different pole orders and hence are linearly independent; their number is $m + 1$, so they form the displayed space's basis.

Every rational function h lies in $k(x)$. To see this without a degree-one-map assertion, list its finitely many poles outside q. At each such a, the function $x - x(a)$ is nonzero and has positive valuation. Multiply h by sufficiently large powers of all these functions to cancel its poles there. The product then lies in some $H^0(\mathcal{O}_C(mq))$, and therefore is a polynomial in x. Dividing by the multiplying polynomial proves the claim. Thus $k(C) = k(x)$. Inverse rational maps between C and $\mathbf{P}^1$ extend at every DVR by their projective homogeneous coordinates and agree globally by the equalizer argument. They are inverse isomorphisms. □

These proofs must be placed before AG-RG-01. In particular, for a nonempty finite boundary divisor D on a smooth completion, $\deg T_C(-D) = 2 - 2g - \deg D < 0$ when $g \geq 1$, and Lemma 6.3

gives $H^0(C, T_C(-D)) = 0$. This is the exact degree-and-genus input used by the one-dimensional unipotent classification; none of it is deferred to a later group lesson.

7. Applying the proof route in the group lessons

The normalizer construction in AG-RG-03 uses Theorem 4.5 to choose smooth affine group and subgroup models with geometrically connected fibres; its finite group equations and closed immersion descend in the same finite list. The projective homogeneous quotient proof uses Theorem 2.3 on its actual projective closure and Corollary 2.4 for the formal idempotent. Lemma 1.3 supplies faithful flatness of local completion, the actual argument of AG-CA-19 Theorem 3.2. Section 5 supplies the line bundle cohomology statement over arbitrary, including nonreduced, bases used in AG-RG-03 and AG-RG-06. Section 6 supplies the smooth curve duality, divisor degrees, genus calculation and domination by a projective line used in AG-RG-01 and AG-RG-03; it must precede AG-RG-01.

For AG-RG-06's use of sufficiently large powers of an ample line bundle, the exact earlier embedding proofs are AG-RG-S01, Lemma 8.1 and Lemma 8.3: finitely many nonvanishing sections with affine domains, denominator clearing for their coordinate generators, and an immersion from those generators. AG-RG-S01, Corollary 7.B turns that immersion into a closed immersion when the source is proper. AG-RG-S01, Theorem 8.5 treats every sufficiently large power by applying its projective finiteness and vanishing argument to all finitely many residue classes modulo the embedding exponent.

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History

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